

Review

The adversarial game between detection and evasion: A survey of anti-detection techniques for machine-generated texts

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ABSTRACT

With the explosive growth of large language models (LLMs), research on machine-generated text detection (MGTD) has also proliferated. Alongside these developments, a wide range of attack algorithms targeting MGTD systems have emerged. While previous studies have surveyed detection techniques, few have examined the dynamic interplay between attack and defense. Following PRISMA 2020, this paper systematically synthesizes 27 studies of attacks against MGTD and the available evidence on corresponding defenses. We categorize existing research into four major types of evasion strategies: watermark attacks, paraphrasing attacks, prompt-based attacks, and adversarial-text attacks, and summarize the available defense evidence. Furthermore, to better understand the practical implications of these methods, we compile the reported performance results of attack and defense techniques across different detectors. Finally, we highlight the current challenges in this area and outline potential future research directions. A companion repository containing the categorized literature, paper links, and available code, data, and project repositories is provided at <https://github.com/AIGC1999/A-Survey-of-Anti-Detection-Techniques-for-Machine-Generated-Texts>.

1. Introduction

In recent years, generative artificial intelligence (AI) models have advanced rapidly. Since the release of ChatGPT by OpenAI in 2022, an increasing number of advanced large language models (LLMs) have emerged, including LLaMA (Touvron et al., 2023), Gemini,¹ and DeepSeek (DeepSeek-AI et al., 2025). These models are trained on vast datasets using extensive computational resources, enabling them to produce human-like text with remarkable fluency. They demonstrate transformative potential across various domains such as natural language processing, automated content creation, and conversational agents. LLMs excel at generating coherent, context-aware, and stylistically diverse outputs, often indistinguishable from human-written content. This capability has facilitated their widespread adoption and driven innovation across a broad spectrum of fields, including healthcare (Li et al., 2024b; Luo et al., 2024), medical decision-making (Li et al., 2024a; Liao et al., 2024), education (Dan et al., 2025), finance (Chen et al., 2023a), and law (Martínez et al., 2025; Yue et al., 2024).

Despite the many benefits of LLMs, their rapid proliferation has also introduced a range of risks, as they are capable of generating highly realistic text that can be maliciously exploited (Ray, 2023; Shahzad et al., 2025). For instance, machine-generated content has been used to spread

misinformation, fake news, and seemingly benign but actually harmful materials, thereby eroding public trust in digital media (Fagni et al., 2021). In education and academia, the misuse of AI to produce essays or research papers poses a serious threat to academic integrity and fairness (Eke, 2023). Furthermore, the automation of phishing emails and social engineering attacks powered by AI presents significant cybersecurity challenges (Emanuela et al., 2024). These risks underscore the urgent need for robust detection technologies capable of distinguishing machine-generated text from human-authored content. Effective detection can help ensure the reliability of information, reduce societal harm, and uphold ethical standards across schools, media outlets, and online platforms. To address the challenges outlined above, researchers have developed a variety of methods for machine-generated text detection (MGTD), encompassing statistical analysis (Mitchell et al., 2023; Su et al., 2023), watermarking (Fernandez et al., 2023; Kirchenbauer et al., 2023; Qu et al., 2025; Ren et al., 2024), and fine-tuned model-based approaches (Ippolito et al., 2020; Liu et al., 2023). Statistical methods, such as GLTR, leverage the probability distributions of language models to distinguish between human-written and machine-generated text (Gehrmann et al., 2019). Watermarking techniques embed traceable signals into generated content, enabling later verification of its origin (Kirchenbauer et al., 2023). Meanwhile, fine-tuned transformer-based

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¹ <https://gemini.google.com/>

models such as ConDA employ deep learning strategies to achieve high detection accuracy on specialized datasets (Bhattacharjee et al., 2023). Although these methods have demonstrated strong performance on carefully curated benchmarks, they remain vulnerable to a growing array of adversarial evasion techniques. These include adversarial rewriting (Krishna et al., 2023), watermark attacks (Chang et al., 2025), and prompt engineering (Koike et al., 2024), all of which are capable of significantly reducing detection accuracy and present ongoing challenges to the robustness of current systems.

The ongoing adversarial dynamics between detection and evasion highlight the necessity of a deeper understanding of both technical paradigms. While several survey studies have extensively reviewed detection algorithms and datasets (Jawahar et al., 2020; Liu et al., 2024b; Tang et al., 2024; Wu et al., 2025; Yang et al., 2025), there remains a lack of systematic exploration of the adversarial interplay between detection and counter-detection strategies. This paper aims to comprehensively organize and analyze attack algorithms and defense strategies in the context of MGTD. We categorize attack methods into four primary types: watermark attacks, paraphrasing attacks, prompt-based attacks, and adversarial text attacks. Watermark attacks are designed to disrupt or erase embedded watermark signals in machine-generated text, rendering it undetectable while preserving its content and intended functionality (Kirchenbauer et al., 2023). Paraphrasing attacks involve using large language models or manual intervention to replace vocabulary, restructure syntax, or reinterpret semantics of machine-generated text, aiming to evade detectors without significantly altering the underlying meaning (Krishna et al., 2023). These methods typically exploit detectors' sensitivity to surface-level features and aim to bypass mechanisms based on statistical distributions, syntactic structure, or feature extraction. Prompt-based attacks manipulate the output of language models through carefully crafted prompts. These include prompt injection, style-shifting instructions, and adversarial prefixes, all intended to interfere with the semantic or stylistic traits of generated outputs at the source, thereby evading detection (Kumarage et al., 2023). Adversarial text attacks reduce the effectiveness of machine-generated text detectors by introducing minimal yet strategically applied perturbations, including spelling alterations, synonym replacements, and irrelevant content insertions. Unlike paraphrasing-based strategies, these attacks generally avoid extensive rewriting and instead focus on subtle manipulations that preserve both the readability and functionality of the text (Wang et al., 2024b).

Given that previous literature has thoroughly analyzed detection algorithms (Jawahar et al., 2020; Liu et al., 2024b; Tang et al., 2024; Wu et al., 2025; Yang et al., 2025), our primary focus is on evasion techniques, with a brief review of detection methods. Specifically, Section 3 provides a concise overview of machine-generated text detection methods, while Sections 5 and 6 systematically classify and evaluate attack and defense strategies against these detectors. Fig. 2 presents the classification of these attack algorithms and the corresponding defense evidence. The systematic evidence synthesis includes studies that propose or empirically evaluate evasion attacks against MGTD and, where available, matched defenses. Studies concerned only with general detection algorithms, without adversarial evaluation, are excluded from the 27-study synthesis; the brief detection overview in Section 3 is contextual and draws on background literature. The contributions and novelty of this study are summarized as follows:

1. This study provides a systematic synthesis focused specifically on the adversarial interplay between detection and counter-detection strategies in MGTD, rather than solely on detection algorithms or datasets.
2. We organize and analyze attack algorithms and defense strategies within the context of MGTD. Specifically, we categorize attack methods into four primary types: watermark attacks, paraphrasing attacks, prompt-based attacks, and adversarial-text attacks, thereby

providing a clear classification framework and adversarial perspective for future research.

2. Literature retrieval methods

The literature search, screening, and study-selection processes followed the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework (Page et al., 2021). The resources and specific procedures used for literature search, screening, and inclusion are detailed below.

2.1. Eligibility criteria

Studies were included in the systematic synthesis if they (1) proposed or empirically evaluated an evasion attack against an MGTD system and/or (2) evaluated a corresponding defense or robustness mechanism. We restricted inclusion to peer-reviewed journal articles, conference papers, and preprints written in English and available in full text. Exclusion criteria were (1) editorials, commentaries, or position papers without empirical validation, (2) duplicate records, (3) studies unrelated to MGTD, and (4) studies concerned only with general detection algorithms without adversarial attack or defense evaluation.

2.2. Information sources

We systematically searched major academic databases and repositories, including ACL Anthology, arXiv, and ACM Digital Library. These sources were selected because they cover the most relevant venues for research on NLP. In particular, ACL Anthology includes leading conferences and journals in computational linguistics, arXiv provides timely access to preprints in computer science and artificial intelligence, and ACM Digital Library indexes peer-reviewed articles in computer science, security, and applied AI. Together, these databases provide broad coverage of the rapidly evolving research on machine-generated text detection, adversarial attacks, and corresponding defense strategies.

2.3. Search strategies

Database-specific search strategies were employed to accommodate the functionalities and limitations of each source. The details are summarized in Table 1. For the ACL Anthology, since the database does not support advanced search functions, we restricted the scope to titles containing terms related to text detection and adversarial methods. Specifically, we entered the keywords "text detection," "AI text detection," "machine-generated text detection," "adversarial attack," "evasion attack," and "adversarial defense" in parallel, which yielded 136 records. For arXiv, we designed a title-focused search strategy because full-text queries returned several thousand records, many of which were irrelevant. We limited the subject area to Computer Science and searched titles for any of the following terms: "machine-generated text," "AI-generated text," "LLM-generated text," "AI text detection," "machine-generated text detection," "adversarial attack," "evasion attack," or "adversarial defense." This strategy resulted in 121 records. For the ACM Digital Library, we conducted a full-text search using the query ("machine-generated text" OR "AI-generated text" OR "LLM-generated text") AND ("AI text detection" OR "machine-generated text detection"). We initially attempted to include adversarial terms such as "adversarial attack," "evasion attack," and "adversarial defense", but this returned zero results. Therefore, we broadened the search scope to focus on detection-related terms only and subsequently performed manual screening. This yielded 48 records.

2.4. Selection process

Our review focused on studies that proposed or empirically evaluated evasion attacks against MGTD and/or evaluated corresponding defenses. The overall selection process is illustrated in the PRISMA flow

Table 1
Search strategies for each academic database.

Database	Search Query	Date	Result
ACL Anthology	query: Title: ("text detection" OR "AI text detection" OR "machine-generated text detection" OR "adversarial attack" OR "evasion attack" OR "adversarial defense")	23 July 2026	136
arXiv	query: Title: ("machine-generated text" OR "AI-generated text" OR "LLM-generated text" OR "AI text detection" OR "machine-generated text detection" OR "adversarial attack" OR "evasion attack" OR "adversarial defense"); filter: Subject: Computer Science	23 July 2026	121
ACM Digital Library	query: Full text: (("machine-generated text" OR "AI-generated text" OR "LLM-generated text") AND ("AI text detection" OR "machine-generated text detection"))	23 July 2026	48

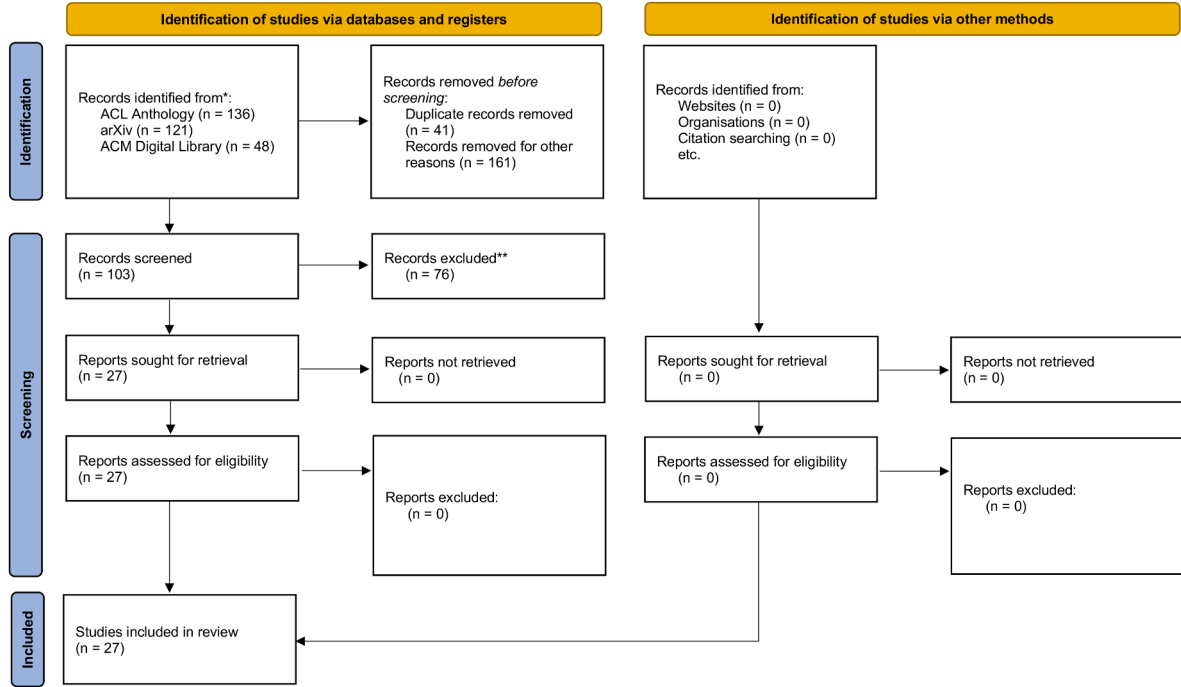


Fig. 1. PRISMA flow diagram for the database searches and study-selection process. The asterisk indicates that database-specific record counts are reported in Table 1. The double asterisk indicates that all 76 exclusions were made through manual title-and-abstract screening; no automation tool made exclusion decisions. Records removed for other reasons were general detection-only studies without adversarial evaluation.

diagram (Fig. 1). The database searches identified 305 records: 136 from ACL Anthology, 121 from arXiv, and 48 from the ACM Digital Library. All records were imported into EndNote for reference management. Before title-and-abstract screening, 41 duplicate records were removed. A further 161 records were removed because their titles and indexing metadata showed that they addressed general detection algorithms only and did not evaluate an adversarial attack or corresponding defense, leaving 103 records for screening. The remaining records were imported into Rayyan.ai, a web-based systematic-review tool that supports blinded screening, conflict resolution, and tagging (Ouzzani et al., 2016). Within Rayyan, records were categorized as excluded, possible, or included. Records categorized as possible underwent a second assessment. Title-and-abstract screening excluded 76 records, leaving 27 reports sought for retrieval. All 27 reports were retrieved and assessed for eligibility, and none was excluded during the full-text assessment. Consequently, 27 studies were included in the final analysis.

2.5. Data extraction and synthesis

For each included study, we extracted bibliographic information, attack category, target detector or model, dataset, threat-model and access assumptions, evaluation metric, reported attack effect, proposed

defense, and empirical defense effect. The values summarized in Table 2 were checked against the full text of each paper. When a study reported multiple experiments, we selected a representative headline result with a clearly specified detector, dataset, metric, and operating point. Matched before→after values were retained when available. If attack and defense values were reported under different detector configurations, we retained the original values but did not interpret them as a matched recovery. Because the included studies differed substantially in datasets, generators, detectors, attack budgets, and evaluation metrics, we used a narrative synthesis rather than a meta-analysis.

3. Brief review of machine-generated text detection algorithms

3.1. What: defining the scope and boundaries

This section defines the scope and boundaries of machine-generated text detection to clarify how these systems operate. Machine-generated text detection determines whether a text sequence was produced by an AI model or a human author. Formally, the task can be defined as a binary classification problem. Given a text sequence T , a detector $D(T)$ outputs a label $y \in \{0, 1\}$, where $y = 1$ indicates machine-generated text

and $y = 0$ indicates human-written text:

$$D(T) = \arg \max_{y \in \{0,1\}} P(y|T, \theta),$$

where $P(y|T, \theta)$ represents the probability of label y given text T and model parameters θ . The scope of detection encompasses diverse text types, generative model characteristics, and granularity levels, necessitating a range of methodologies to address varying contexts and challenges. In the following subsections, we briefly review some commonly used detection algorithms, providing only a concise overview. Because the primary scope of this review is the adversarial interaction between attacks and defenses, this section is not intended to provide an exhaustive survey of general MGTD algorithms. Instead, it introduces the principal detection paradigms and the evidence on which they rely, thereby establishing the technical targets whose vulnerabilities and countermeasures are examined in the subsequent sections.

3.2. Training-based algorithms

Supervised detection methods rely on training models with labeled datasets that contain both machine-generated and human-authored text. These models aim to learn discriminative features between the two classes, achieving high performance across diverse scenarios. However, this approach requires carefully designed feature engineering and robust model architectures to handle various generation mechanisms and adversarial manipulations.

Ippolito et al. (2020) fine-tuned a BERT model to build an automated detector for distinguishing between human-written and machine-generated text. Their dataset included texts generated by GPT-2 using top- k , nucleus sampling, and random decoding strategies, as well as human-written counterparts. BERT performed particularly well on long-form text, and among the decoding strategies, top- k sampling yielded outputs with more statistically detectable anomalies. Nonetheless, the model exhibited limited generalizability across different decoding strategies, and detection accuracy was highly sensitive to text length. To mitigate this sensitivity, Tian et al. (2024) proposed a Multi-scale Positive-Unlabeled (MPU) detection framework. Treating short machine-generated texts as partially unlabeled data, MPU employs a length-aware loss function and multi-scale augmentation to dynamically adjust priors based on input length. This significantly improves detection performance for shorter texts. Wang et al. (2023) introduced SeqXGPT, a sentence-level machine-generated text detector capable of both sentence- and document-level classification. The model interprets a list of token-level log-probabilities as waveform-like input features, processes them with a five-layer CNN for local pattern extraction, and then applies two Transformer layers to capture long-range dependencies.

A linear classifier assigns token-wise labels, and majority voting is used to determine the final class label. Similarly, Chen et al. (2023b) used a frozen RoBERTa-base encoder with a fine-tuned MLP classifier and represented each input using the hidden state of the [CLS] token. Although this method achieved strong results on its proposed dataset, it showed poor generalization to unseen generators, highlighting generator-specific biases in detection performance. To enhance generalizability, Bakhtin et al. (2019) used pre-trained autoregressive language models to synthesize negative samples for energy-based detection, increasing both generator diversity and data volume. Empirical results indicate that larger models and more extensive training data significantly improve detection accuracy. In the absence of labeled in-domain data, Bhattacharjee et al. (2023) proposed ConDA, an unsupervised contrastive domain adaptation framework that achieves near-supervised performance. ConDA constructs a twin-network architecture using RoBERTa-base and jointly optimizes classification, contrastive, and domain discrepancy losses. It demonstrates substantial improvements across multiple datasets, even under label-scarce target domains.

3.3. Zero-shot algorithms

Zero-shot algorithms detect machine-generated text without requiring labeled training data, relying instead on inherent properties of the text or model outputs. These methods are particularly valuable for rapid deployment, handling novel generative models, or scenarios with limited labeled data. Prior to the release of ChatGPT, GLTR (Gehrmann et al., 2019) was a prominent zero-shot detection algorithm designed for identifying text generated by pre-trained language models. GLTR is a visualization-based tool that exploits the statistical properties of language models to highlight anomalies in token distributions that are indicative of machine-generated content. In addition to GLTR, several statistical methods have been proposed, including DetectGPT (Mitchell et al., 2023) and Fast-DetectGPT (Bao et al., 2024). DetectGPT is a zero-shot detection approach based on probability curvature. It detects machine-generated text by observing that such content often lies in regions of negative curvature in the log-probability function. By generating perturbed versions of a given text using the same pre-trained model and analyzing log-probability differences, DetectGPT can effectively identify machine-generated text. Fast-DetectGPT improves upon DetectGPT by employing conditional probability curvature, which captures contextual differences in word selection between human and machine. This significantly enhances both detection accuracy and computational efficiency.

Further advancing this line of work, Su et al. (2023) proposed a method based on normalized perturbation log-rank. Their approach not only requires fewer perturbations than DetectGPT but also achieves better performance under equivalent conditions. Another strategy involves Bayesian surrogate models, which significantly reduce the number of queries needed for detection (Miao et al., 2024). By using a Gaussian process as the surrogate and leveraging Bayesian uncertainty to select representative samples, this method interpolates detection scores across other instances, thus improving query efficiency while maintaining detection accuracy. In addition, machine-generated text typically exhibits lower perplexity than human-authored text. HowkGPT (Vasilatos et al., 2025) exploits this characteristic by comparing perplexity scores of student-written and ChatGPT-generated content through context-aware analysis. This lightweight, web-deployable tool shows promise for academic-integrity applications that require suspicious submissions to be flagged.

3.4. Watermark-based detection algorithms

Watermark-based detection algorithms involve embedding identifiable signals into text during the generation process, enabling reliable identification when model internals are accessible. Unlike training-based or zero-shot methods, watermarking is a proactive approach integrated into the generative model's design, typically used in white-box settings. For instance, Ren et al. (2024) proposed SemaMark, which maps the semantics of the preceding context to a region of an embedding space and uses the resulting region to guide token selection. By tying the watermark signal to contextual meaning rather than a fixed surface-token partition, SemaMark is designed to retain detectable evidence after paraphrasing. Liu et al. (2024a) further introduced UPV, an unforgeable publicly verifiable watermarking scheme. It uses two independent neural networks for watermark generation and detection, thus eliminating security risks associated with traditional shared-key designs.

Generated-text watermarking should be distinguished from model-ownership watermarking. Distillation-resistant and backdoor-based schemes embed verifiable behavior in a model to protect intellectual property or establish ownership (Gu et al., 2023; Zhao et al., 2022). They do not directly determine whether a particular text was machine generated and are therefore not treated as MGTD algorithms in this review.

4. Definition of attack algorithms and evaluation metrics

4.1. What: defining attack algorithms

Attack algorithms, often referred to as anti-detection or adversarial strategies, aim to evade or undermine detection systems designed to identify LLM-generated text. The primary objective is to modify or generate text such that it appears indistinguishable from human-written text, thereby bypassing detectors and enabling misuse in applications such as misinformation, academic fraud, or automated phishing. Formally, the goal of an attack algorithm can be defined as an optimization problem. Given a text sequence T generated by an LLM and a detector $D(T)$ that outputs a label $y \in \{0, 1\}$, the attacker seeks to produce a modified text T' such that:

$$T' = \arg \min_{T'} P(y = 1 | T', \theta_D),$$

where $P(y = 1 | T', \theta_D)$ is the probability that the detector D with parameters θ_D classifies T' as machine-generated, and T' is constrained to preserve the semantic meaning or intent of T . Attack algorithms exploit vulnerabilities in detection systems, such as reliance on specific features or lack of robustness to adversarial modifications, thereby creating an ongoing adversarial interaction between attacks and detectors.

4.2. Evaluation metrics

To properly evaluate the effectiveness of attacks and the robustness of detection systems for identifying LLM-generated text, a wide range of metrics are employed. Each of these metrics captures different dimensions of detector performance, and selecting the right combination is essential for comprehensive evaluation. Below, we describe the key metrics used in recent studies and explain their significance. False-negative rate (FNR) measures the proportion of machine-generated texts that are mistakenly classified as human-written. A high FNR indicates that the detector is failing to catch generated content. Attack success rate (ASR) refers to the percentage of adversarial samples that successfully bypass the detection system. True-positive rate at an $X\%$ false-positive rate (TPR@X%FPR) quantifies how well a detector identifies positive samples when constrained to a very low false alarm rate. It is a key metric in evaluating watermarked text detectors (Chang et al., 2025; He et al., 2024). Area under the receiver operating characteristic curve (AUROC) evaluates a detector's overall ability to discriminate between human and machine-generated text. It is commonly used in adversarial evaluation; drops in AUROC reflect degraded performance under attack (Artemova et al., 2025; Wang et al., 2024b). Accuracy (ACC) measures the overall correctness of predictions on MGTD. The F1 score is the harmonic mean of precision and recall and is useful when positive-class precision and recall are the main evaluation priorities. However, it does not account for true negatives and should not be interpreted as a generally imbalance-robust metric. A significant decline in F1 is often used to demonstrate the success of an attack (Kumarage et al., 2023; Tripto et al., 2024). Macro F1 treats each class equally, averaging F1 scores across classes. It is particularly useful for multi-class detection or severely imbalanced binary tasks (Tripto et al., 2024). Z-score, used in statistical watermark detection, indicates how distinguishable the generated text is from human-written text based on language features. Lower Z-scores suggest successful evasion (Chakraborty et al., 2023; Zhang et al., 2024a). A p -value is often paired with a z-score to quantify how compatible the observed watermark statistic is with the null hypothesis. A high p -value indicates insufficient evidence to reject the null hypothesis at the selected significance threshold; it does not prove that machine- and human-generated texts are equivalent or inherently inseparable. Mean Average Precision (mAP) is used in ranking-based or retrieval-style detection settings. Lower mAP values under attack demonstrate that detectors struggle to rank machine-generated text correctly (Zhang et al., 2024b). Overall, these metrics offer complementary perspectives

on how detection models perform under different types of adversarial attacks.

5. Attack algorithms against machine-generated text detection

5.1. Watermark attacks

Watermark attacks target signals inserted during generation rather than features learned by an external classifier. Distribution-modifying schemes partition the vocabulary into preferred and non-preferred tokens, then bias sampling toward the preferred set (Kirchenbauer et al., 2023). Detection usually tests whether the observed token sequence contains an unexpectedly large proportion of preferred tokens. An attacker therefore succeeds by estimating the hidden partition, reducing its statistical signal, or transforming the text across a representation where the partition is not preserved. The relevant threat models range from black-box access to generated samples to public detector scores and model token probabilities.

Watermark stealing treats repeated outputs from a watermarked model as observations of its hidden sampling rule (Jovanović et al., 2024). The attacker queries the model under varied contexts and estimates how strongly candidate tokens are favored by the watermark. This estimated scoring function supports two distinct attacks. In spoofing, an auxiliary generator favors tokens predicted to carry the target watermark, causing unrelated text to be falsely attributed to the protected model. In scrubbing, the same estimate guides a paraphraser away from high-scoring tokens while retaining fluent alternatives. For KGW2-SelfHash, stealing-assisted scrubbing raises quality-filtered FNR at 10^{-3} from 0.00 to 0.79 on Reddit Writing. On Dolly-CW, the corresponding change is from 0.02 to 0.90. The study also shows that practical stealing can be mounted through paid API queries at modest cost, making the attack relevant beyond a purely theoretical setting.

Other attacks exploit design choices that expose token-frequency or confidence information. Pang et al. (2024) show that using multiple watermark keys creates frequency patterns that can be estimated through repeated queries. Preferentially sampling tokens inferred to have low watermark confidence can remove the signal, whereas mimicking high-confidence tokens can produce spoofed text. When a public detector API exposes confidence scores, candidate substitutions can be selected directly against the returned score. In the reported KGW experiment, removal and spoofing reach ASRs of 1.00 and 0.98, respectively. Smoothing uses a different signal: the correlation between model confidence and a token's watermark contribution (Chang et al., 2025). It selectively re-samples low-confidence positions using either a reference model or the target distribution. Across ten watermarking methods and models from 1.3B to 30B parameters, this procedure removes substantial watermark evidence while preserving perplexity and diversity. For KGW on OPT-1.3B, TPR falls from 100% to 0% at an FPR below 1%.

Color-aware and oracle-guided attacks require less direct knowledge of the watermark rule. Self-Color Testing-based Substitution (SCTS) prompts the watermarked model repeatedly and compares token frequencies to infer whether candidate tokens behave as preferred or non-preferred colors (Wu & Chandrasekaran, 2024). It then replaces inferred preferred tokens with semantically compatible alternatives of the opposite color. Because the attack targets the watermark contribution of each edit, it can remain effective on long passages where color-agnostic paraphrasing only dilutes the signal. In the reported UMD configuration, AUROC decreases from 0.9884 to 0.1332. The oracle-guided framework of Zhang et al. (2024a) is more general and does not assume a particular watermark family. A perturbation oracle proposes local changes, while a quality oracle rejects candidates that degrade the response. Repeated accepted edits form a random walk over high-quality outputs until the watermark statistic becomes insignificant. For Unigram, the z-score decreases from 8.210 to 1.456, and the p -value increases to 0.208.

Cross-lingual and hybrid transformations attack the stability of watermark features across representations. Cross-lingual Watermark Re-

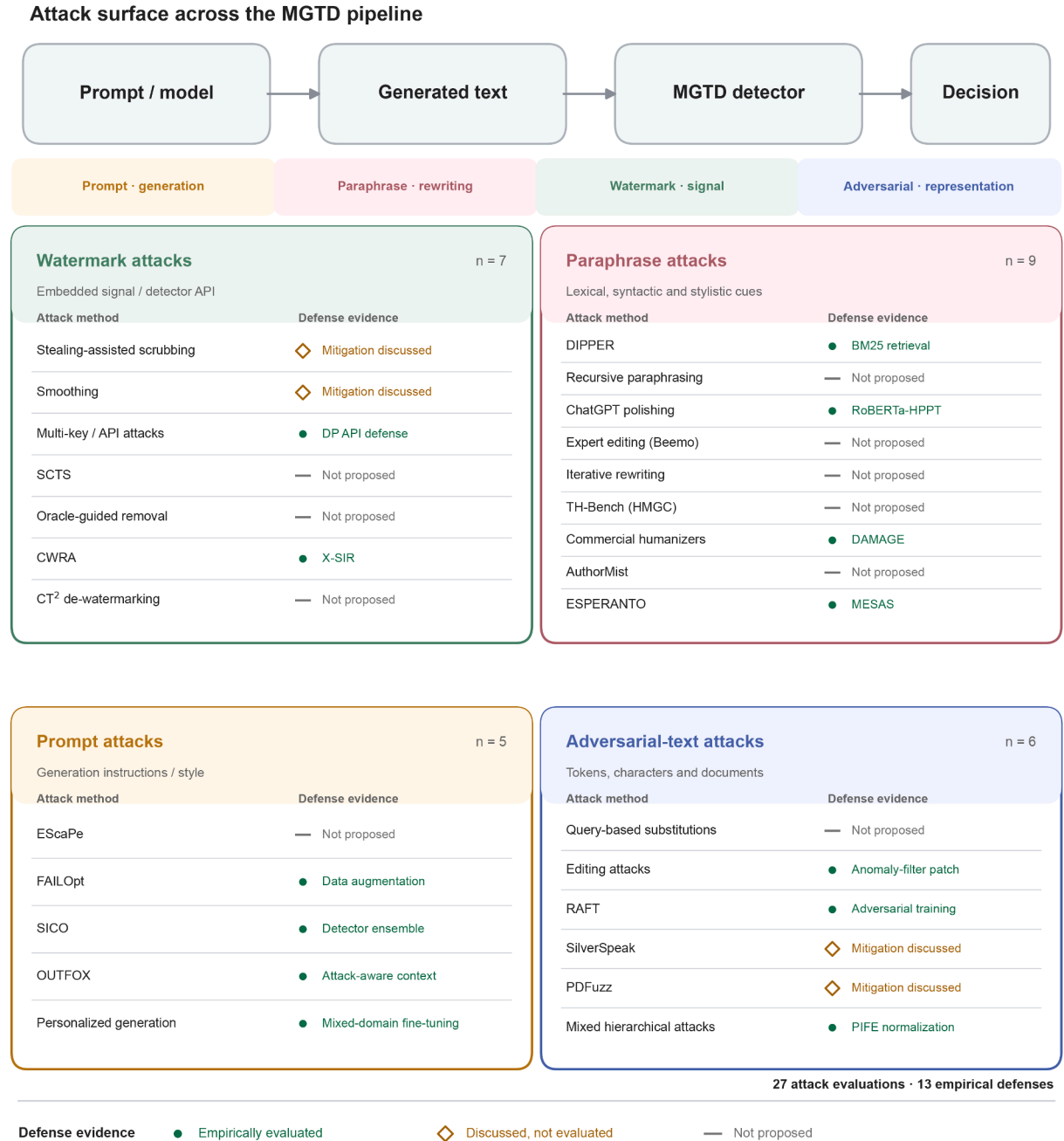


Fig. 2. Attack surfaces and defense evidence for the methods summarized in Table 2. Attacks are organized into watermark, paraphrase, prompt, and adversarial-text categories according to the generation or detection component they target. Circles indicate empirically evaluated defenses, diamonds indicate mitigation ideas without empirical defense results, and dashes indicate that no defense method was proposed in the cited study.

removal Attack (CWRA) asks the model to respond in a pivot language and translates the response into the target language (He et al., 2024). Translation preserves the response meaning but changes tokenization, word order, and lexical choices that carry the original watermark. In the reported KGW setting, TPR at 10% FPR decreases from 0.992 to 0.213. CT² combines high-entropy token replacement with semantic paraphrasing (Chakraborty et al., 2023). The first stage replaces uncertain content words with semantically similar alternatives, and the second stage restructures the passage with a paraphrasing model. Across the evaluated paraphraser, de-watermarking accuracy reaches 99.2%, while GPT-3.5 reaches 99.1%. Because its target is an embedded watermark signal, CT² is classified here as a watermark attack rather than a general adversarial-text attack.

5.2. Paraphrasing attacks

Paraphrasing attacks preserve the main proposition of a passage while changing the lexical, syntactic, and discourse features used by detectors. DIPPER operationalizes this threat with an 11B-parameter T5-XXL paraphraser that conditions on surrounding context (Krishna et al., 2023). Two control codes regulate lexical diversity and content reordering, allowing the attacker to vary perturbation strength without changing the task or prompt. The resulting text evades watermark detectors, DetectGPT, GPTZero, and the former OpenAI classifier. On GPT-2-XL outputs, DetectGPT's detection rate falls from 70.3% to 4.6% at a 1% FPR. The attack remains semantically close to the source, although stronger controls introduce a quality-evasion trade-off.

Recursive paraphrasing strengthens this mechanism by applying several rounds of rewriting rather than a single transformation (Sadasivan et al., 2025). Each round moves the text farther from the generator's original token distribution and watermark pattern. In a detector-aware setting, multiple candidates can be generated at each round and ranked using the target detector's score. This selection step converts a generic paraphraser into an adaptive black-box attack. The study evaluates approximately 300-token passages across watermarking, neural, zero-shot, and retrieval-based detectors. For the evaluated watermark detector, TPR at 1% FPR decreases from 99.3% to 9.7%. Text quality declines gradually with additional rounds, so attack strength cannot be interpreted independently of semantic preservation.

Human and model editing introduce paraphrase distributions that differ from fully automated attacks. ChatGPT polishing starts from human-written academic abstracts and applies instructions that improve grammar, fluency, and presentation (Yang et al., 2023). The edited text can acquire features associated with model generation, creating false positives rather than only hiding machine-generated content. The authors construct the HPPT dataset of paired original and polished abstracts and introduce Polish Ratio to quantify the extent of model intervention. Beemo studies a complementary setting in which 25 professional editors revise machine-generated text for flow, coherence, and factual quality (Artemova et al., 2025). Its 6.5k core texts and 13.1k model-edited variants cover multiple tasks and ten instruction-tuned models. Expert editing reduces detector AUROC by up to approximately 22 percentage points, whereas model-only editing is less consistently perceived as human.

Iterative style transfer also complicates the meaning of authorship. The Ship of Theseus study repeatedly paraphrases a passage and measures how classification changes as the surface realization drifts from the original author (Tripto et al., 2024). The attack does not only seek a human label; it progressively replaces stylistic evidence associated with the source. On ELI5, Macro F1 for a fine-tuned BERT detector decreases from 0.96 to 0.51 after ChatGPT paraphrasing. This result illustrates that detector failure and authorship ambiguity can arise together, particularly when a document is revised through many model-assisted iterations.

TH-Bench provides a common evaluation framework for comparing such humanization attacks (Zheng et al., 2025). It evaluates six attacks against 13 detectors on six datasets, covering 19 domains and outputs from 11 language models. The benchmark separates three objectives: evasion effectiveness, preservation of text quality, and computational overhead. This design exposes trade-offs that are hidden when an attack is judged only by detector failure. No method dominates all three dimensions. As one representative result, the Log-Likelihood detector's AUROC on Reuters decreases from 0.808 to 0.020 under human-machine collaborative generation. TH-Bench is therefore included as an attack benchmark rather than as a new defense method.

Commercial humanizers broaden the threat model because they are accessible without model weights or detector internals. DAMAGE evaluates 19 humanization and paraphrasing services, examining both detector evasion and faithfulness to the source (Masroue et al., 2025). Many services substantially weaken existing detectors, although their semantic preservation and rewriting behavior vary. Under humanization, the TPR of Binoculars at 5% FPR decreases from 0.9415 to 0.2823. The authors also train a robust detector through data-centric augmentation with outputs from diverse humanizers. This detector generalizes to held-out tools and retains a TPR of 0.9826 on humanized academic text. They further attack their own detector with a model optimized against its predictions, testing whether cross-humanizer diversity provides resistance beyond memorized services.

AuthorMist makes detector feedback part of the paraphrasing objective (David & Gervais, 2025). A 3B-parameter language model is optimized with group-relative policy optimization, while scores from external detector APIs act as rewards. The same framework can target GPTZero, Winston AI, Originality.ai, or any detector that exposes

a usable prediction interface. The reward encourages low detectability while semantic constraints preserve the source meaning. Reported semantic similarity remains above 0.94, and individual-detector ASRs range from 78.6% to 96.2%. The strongest cross-detector configuration reaches a mean ASR of 95.17%, showing that detector-specific feedback can transfer across several commercial systems.

ESPERANTO treats multilingual back-translation as a paraphrasing operator rather than a watermark-specific transformation (Ayooobi et al., 2025). AI-generated text is translated through one or more intermediate languages and then returned to English. Candidate back-translations can be combined to preserve meaning while replacing detector-sensitive lexical and syntactic patterns. Evaluation across six open-source and three proprietary detectors shows substantial vulnerability. For example, GPTZero TPR at 1% FPR on the R-L setting decreases from 0.65 to 0.09. The paper also introduces MESAS, which ranks unigram and bigram features using a modified separation score before training a TF-IDF-based classifier. MESAS retains a mean post-attack TPR of 0.947, with only a 1.85% relative decline under back-translation.

5.3. Prompt-based adversarial attack strategies

Prompt-based attacks intervene before text is generated. Instead of rewriting a completed passage, they condition the generator to avoid features associated with machine-generated text. Evasive Soft Prompt (EScaPe) learns a continuous prompt whose objective is to maximize the probability that a detector assigns a human label (Kumarage et al., 2023). The method first optimizes virtual prompt embeddings for a source language model using proximal policy optimization. A lightweight projection module then transfers the learned prompt to other models with different embedding spaces. This two-stage design separates attack optimization from the final generator and supports cross-model evaluation. Across the reported tasks and models, mean F1 for the OpenAI fine-tuned detector decreases from 0.954 to 0.537. Because the prompt is continuous, the attacker requires more access than a user limited to ordinary natural-language instructions.

FAILOpt targets prompt-specific shortcuts introduced when detector training corpora are generated from a narrow set of instructions (Park et al., 2024). The attack begins by generating candidate instruction terms that may alter style, discourse structure, or task realization. It filters ineffective candidates and iteratively updates the list using detector feedback. The optimized instruction avoids patterns that the detector has associated with the original collection prompts rather than with machine authorship itself. In the reported SQuAD setting, ChatGPT-detector AUROC decreases from 0.9184 to 0.4452. The maximum ASR reaches 95.72%. Statistical detectors such as DetectGPT are less consistently affected because their scores do not rely on the same supervised prompt shortcuts.

Substitution-based In-Context Example Optimization (SICO) uses discrete demonstrations instead of continuous prompt vectors (Lu et al., 2024). It constructs in-context examples from human and machine texts, identifies linguistic differences, and repeatedly substitutes synonyms or paraphrases grammatical structures. A proxy detector scores each revision, allowing the examples to evolve toward lower detectability. SICO-Gen conditions new generation, whereas SICO-Para rewrites an existing passage. This separation allows the framework to test whether evasion occurs at generation time or during post-editing. On the writing task, SICO reduces Log-Rank AUROC from 0.914 to 0.147. Human evaluation indicates that the optimized outputs retain readability and task completion, so the reduction is not explained only by visibly corrupted text.

OUTFOX models attack and detection as an iterative in-context interaction (Koike et al., 2024). The attacker receives detector labels and places them alongside task instructions and example essays. It then generates outputs that resemble examples labeled as human. The detector performs the reverse operation by adding adversarial essays and their labels to its own in-context demonstrations. This setting is stronger than static prompting because both sides adapt to the other's recent output

without updating model weights. Against the HC3 detector, attacker-generated essays reduce F1 from 0.738 to 0.007. The attack-aware OUT-FOX detector raises F1 from 0.394 to 0.807, demonstrating that carefully selected adversarial context can function as an inference-time defense.

Personalized generation creates a related distribution shift even when evasion is not the user's explicit objective (Gao et al., 2026). StyloBench pairs literary and blog texts with model-generated imitations of individual writing styles. Detectors that rely on diversity, coherence, or likelihood cues can fail because personalization reverses the direction of these features between human and machine text. The authors describe this phenomenon as the feature-inversion trap. Mean AUROC falls to 0.3233 on Stylo-Literary, while Fast-DetectGPT reaches 0.0871. StyloCheck perturbs texts along candidate stylistic features and measures detector dependence, predicting cross-domain performance shifts with a correlation of 0.85. Mixed-domain RoBERTa fine-tuning recovers 99.95% accuracy on Stylo-Literary but only 71.94% on Stylo-Blog, revealing limited transfer across personalized domains.

5.4. Adversarial-text attacks

Adversarial-text attacks search for small, targeted perturbations that change detector decisions while preserving meaning, readability, or visual appearance. The red-teaming framework of Shi et al. (2024) studies attacks at both generation and output levels. For output attacks, an auxiliary language model proposes contextually appropriate synonyms for influential tokens. A query-based evolutionary search then selects replacement positions and candidates that minimize the detector score. For generation-time attacks, the auxiliary model searches for style or template instructions that shift the output away from detector-sensitive cues. The threat model also considers an auxiliary model whose outputs may themselves be monitored. On GPT-2-XL/XSum, query-based substitution reduces DetectGPT AUROC from 0.844 to 0.039, while the corresponding RoBERTa ASR reaches 68%.

Stumbling Blocks evaluates robustness across editing, paraphrasing, prompting, and co-generation attacks under explicit perturbation budgets (Wang et al., 2024b). Character-level operations include typos and symbol changes, while word- and phrase-level operations alter lexical or syntactic structure. Co-generation attacks insert temporary emojis or spelling noise during decoding and remove them from the final output, disrupting detectors that depend on generation trajectories. The study finds that no evaluated detector remains robust across all attack families. Averaged across detectors and attacks, performance decreases by approximately 35%. For DetectGPT-10z, typo insertion reduces AUROC from 0.8516 to 0.2735. The authors trace several failures to abnormal tokenization and unstable likelihood estimates, then test lightweight pre-processing and anomaly-filter patches.

Replacement-based Adversarial Fine-Tuning (RAFT) is a grammar-preserving black-box substitution attack (Wang et al., 2024a). It uses a surrogate model to estimate token importance and exploits transferability in contextual embedding spaces to identify candidate positions. GPT-3.5 then generates semantically compatible replacements, and candidates are selected greedily against the target detector. This procedure avoids the conspicuous grammatical errors produced by character noise and maintains high semantic similarity in human evaluation. The attack transfers across generators, detectors, and domains. On XSum, Fast-DetectGPT AUROC decreases from 0.9903 to 0.0006, and TPR at 5% FPR decreases from 0.94 to 0. RAFT examples can also be reused for adversarial training, linking attack generation directly to a matched defense.

SilverSpeak attacks the text representation below the word level through Unicode homoglyphs (Creo & Pudasaini, 2025). Visually similar characters from different scripts, such as Latin and Cyrillic forms, are substituted without substantially changing what a reader sees. The perturbation affects tokenization, likelihood computation, and input normalization differently across detector architectures. The study evaluates seven detectors on five datasets, including trained classifiers, statistical detectors, and watermarking methods. Some systems collapse toward a

single predicted class rather than merely losing a few points of accuracy. Mean MCC decreases from 0.64 to -0.01 . Character-set restrictions and Unicode normalization are discussed as mitigations, but no universal or empirically validated defense is established.

PDFuzz exploits the difference between a PDF's visual layout and the character order returned by text-extraction software (Creo, 2025). It changes character positioning and internal object order so that the page appears unchanged to a human reader while the detector receives a scrambled sequence. The attack therefore requires neither paraphrasing nor modification of the visible wording. Against ArguGPT, accuracy decreases from 0.936 to 0.504, F1 decreases from 0.938 to 0, and TPR at 1% FPR decreases from 0.734 to 0.002. This result identifies the document parser as part of the detector's attack surface. Text normalization and structural PDF analysis are proposed as possible mitigations, but their defense effect is not evaluated.

Perturbation-Invariant Feature Engineering (PIFE) treats the magnitude and type of an attack as detection evidence (Teja et al., 2025). A multi-stage normalization pipeline maps perturbed text toward a canonical form. The framework then measures how much normalization was required using edit distance, semantic similarity, and related perturbation features. These signals are supplied to the classifier alongside the normalized representation. Evaluation spans character-, word-, and sentence-level attacks, including mixed settings. Standard adversarial training remains effective against simple surface noise but crosses a semantic evasion threshold under stronger sentence-level transformations. Under sentence-level mixed attacks, its TPR at 1% FPR is 0.488. PIFE raises TPR to 0.826 and AUROC from 0.921 to 0.974, showing the benefit of explicitly modeling the perturbation process.

6. Defense algorithms for machine-generated text detection

6.1. Defenses against watermark attacks

Watermark defenses can intervene at generation, detector access, or representation design. API-level defenses are most relevant when an attack depends on repeated confidence queries. Pang et al. (2024) propose adding input-dependent pseudo-random noise to public detector scores using ideas from differential privacy. Unlike independent random noise, the same input receives a consistent perturbation, preventing an attacker from averaging repeated queries to recover the original score. In the KGW experiment, the mechanism reduces detector-API spoofing ASR from 0.98 to 0, while detector accuracy changes from 0.982 to 0.972. The defense is less effective against removal because an attacker can still search for edits that reduce the watermark without reproducing its distribution. Rate limiting, authentication, and restricted score precision are therefore complementary system controls rather than substitutes for a robust watermark.

Generation-side mitigations attempt to reduce the information that an attacker can infer. Watermark stealing may be made harder by rotating keys, reducing repeated contexts, or using sampling-modification schemes whose behavior is less directly visible in token frequencies (Jovanović et al., 2024). These options introduce trade-offs. More keys can raise the false-positive rate or weaken the evidence accumulated by each detector, while hiding model probabilities conflicts with transparency and debugging requirements. Smoothing similarly motivates restricting access to top- K probabilities because the attack estimates confidence from the exposed distribution (Chang et al., 2025). The paper discusses this defense but does not report a quantitative recovery result. SCTS, oracle-guided removal, and CT² likewise provide attack results without an empirically evaluated countermeasure in the corresponding settings (Chakraborty et al., 2023; Wu & Chandrasekaran, 2024; Zhang et al., 2024a).

Cross-lingual attacks require a watermark whose signal is stable under translation rather than only under same-language edits. X-SIR addresses this requirement by improving semantic consistency across languages (He et al., 2024). Its design uses cross-lingual semantic clustering

to align token or phrase representations and prefix-segmentation consistency to reduce language-specific boundary effects. Under the evaluated cross-lingual settings, X-SIR improves mean TPR by 0.448 and AUROC by 0.221. The gain is substantial but depends on multilingual semantic resources and vocabulary coverage. It does not establish robustness against stealing, confidence-guided smoothing, or oracle-guided editing, which exploit different interfaces.

6.2. Defenses against paraphrasing-based evasion attacks

A retrieval-based defense replaces distributional authorship inference with provenance matching (Krishna et al., 2023). An API provider stores previously generated outputs and searches this database when a candidate passage is submitted. Exact matching handles unmodified outputs, while BM25 or semantic retrieval can recover a source after paraphrasing. On the strongest GPT-2-XL DIPPER setting, BM25 retrieves 97.8% of paraphrases at a 1% FPR. The method avoids relying on fragile likelihood patterns, but its coverage is limited to outputs retained by the provider. Short passages provide fewer retrieval cues, and storing generations raises privacy, access-control, and database-scale concerns. Recursive paraphrasing can also weaken retrieval when successive edits remove distinctive spans.

Attack-aware training learns from the kinds of edits expected at deployment. For ChatGPT polishing, the HPPT dataset pairs human-written academic abstracts with their polished versions (Yang et al., 2023). RoBERTa-HPPT learns features associated with model intervention rather than only fully generated text. It reaches an accuracy of 0.8825 and an AUROC of 0.9518 on the out-of-domain CDB dataset. Polish Ratio provides an accompanying estimate of how extensively the model changed the source. DAMAGE extends this strategy to commercial humanizers by collecting outputs from diverse tools and augmenting the detector's training data (Masrour et al., 2025). Diversity across humanizers is central because a detector trained on one rewriting service may learn service-specific artifacts. The resulting detector retains a TPR of 0.9826 on humanized academic text and remains robust when attacked by a separately optimized paraphraser.

Feature aggregation offers a complementary defense against multilingual paraphrasing. MESAS compares how unigram and bigram entities separate human and machine text before and after back-translation (Ayoobi et al., 2025). It ranks entities with a modified separation score, selects stable features, and trains logistic regression on TF-IDF representations. Combining unigram and bigram evidence reduces dependence on individual lexical choices introduced by translation. Across the evaluated intermediate languages and datasets, mean post-attack TPR remains 0.947 at a 1% FPR. However, the result addresses back-translation specifically. TH-Bench and AuthorMist report strong humanization attacks without matched recovery defenses, while Beemo shows that expert editing remains difficult because it can remove model artifacts without introducing a consistent automated signature (Artemova et al., 2025; David & Gervais, 2025; Zheng et al., 2025).

6.3. Defenses against prompt-based attacks

Prompt-attack defenses primarily expand the detector's training or inference context with adversarially varied instructions. FAILOpt uses the attack itself to discover prompts that expose shortcut dependence (Park et al., 2024). Outputs generated from these prompts are added to the training set so that the classifier cannot rely on the original collection instructions. In the reported SQuAD setting, augmentation raises ChatGPT-detector AUROC from 0.4452 under attack to 0.9888. Improvements also transfer across several tasks, generators, and prompt variants, although the defense still depends on the diversity of optimized prompts used for augmentation. EScaPe does not provide a matched empirical defense, and continuous soft prompts may explore directions that are difficult to cover with discrete instruction augmentation (Kumarage et al., 2023).

SICO combines detectors that respond to different statistical and learned signals (Lu et al., 2024). The ensemble is trained with original and SICO-generated samples, allowing one component to compensate when the other becomes unreliable under optimized context. On writing/SICO-Gen, AUROC increases from 0.099 to 0.691. The recovery is meaningful but remains below clean-setting performance, indicating that ensembling reduces rather than eliminates the attack advantage. Continuous collection of new adversarial samples may also be required as the prompt optimizer changes its linguistic strategy.

OUTFOX implements defense at inference time through dynamic in-context examples (Koike et al., 2024). The detector prompt includes human essays, original model essays, and adversarial essays generated against earlier detector decisions. It can therefore adapt its decision boundary without gradient-based retraining. In the matched adversarial setting, F1 increases from 0.394 to 0.807. Personalized generation requires a different response because its failure arises from cross-domain feature inversion rather than a single optimized prompt (Gao et al., 2026). Mixed-domain RoBERTa fine-tuning reaches 99.95% accuracy on Stylo-Literary but only 71.94% on Stylo-Blog. StyloCheck helps identify unstable features, yet the remaining cross-domain gap shows that diagnostic accuracy does not itself guarantee robust detection.

6.4. Defenses against adversarial-text attacks

Input filtering is effective when an attack creates detectable preprocessing anomalies. Stumbling Blocks proposes patches tailored to the observed failure mechanism (Wang et al., 2024b). For typo attacks, anomalous tokens or unstable likelihood contributions are filtered before the DetectGPT score is aggregated. This raises DetectGPT-10z AUROC from 0.2735 to 0.6032. The partial recovery shows that filtering can suppress obvious noise, but it also removes useful evidence and does not restore the clean AUROC of 0.8516. More structured paraphrases and co-generation attacks may not produce local anomalies that can be removed safely.

Adversarial training exposes a classifier to realistic substitutions before deployment. RAFT-generated examples preserve grammar and semantics, making them more informative than synthetic character noise (Wang et al., 2024a). Fine-tuning Raidar on word-swapped samples raises attack AUROC on XSum from 0.6000 to 0.7333. The same training can reduce clean performance, and robustness transfers imperfectly to different replacement policies or domains. This limitation follows from the attack-specific nature of the augmented distribution. Maintaining protection therefore requires new samples as surrogate models, generators, and optimization objectives evolve.

PIFE reduces dependence on a fixed attack catalog by explicitly measuring the transformation needed to normalize an input (Teja et al., 2025). Character normalization, lexical canonicalization, and sentence-level processing produce a standardized representation. Edit distance and semantic similarity then quantify how far the observed text lies from that representation. Under sentence-level mixed attacks, PIFE raises TPR at 1% FPR from 0.488 for adversarially trained ModernBERT to 0.826, while AUROC increases from 0.921 to 0.974. This approach remains dependent on the normalization pipeline and may miss perturbations that preserve its canonical features. SilverSpeak suggests character-set constraints, Unicode normalization, or OCR-based processing, whereas PDFuzz suggests text normalization and PDF-structure analysis (Creo, 2025; Creo & Pudasaini, 2025). Neither study reports an empirical defense effect, so these suggestions remain open mitigation directions rather than validated countermeasures.

6.5. Effectiveness analysis of attacks and defenses

As shown in Table 2, the reported effects are highly dependent on the detector, dataset, operating point, and threat model; consequently, raw before-after values are preferable to cross-paper percentage comparisons. Watermark attacks can be particularly damaging: smoothing

Table 2
Reported effects of attacks on machine-generated-text detectors and of evaluated countermeasures.

Category	Study / method	Metric	Reported attack effect	Reported defense effect
Watermark	Stealing-assisted scrubbing (Jovanović et al., 2024)	FNR* @10 ⁻³	KGW2-SelfHash: Reddit Writing, 0.00 → 0.79; Dolly-CW, 0.02 → 0.90.	Not evaluated.
Watermark	Smoothing (Chang et al., 2025)	TPR at FPR < 1%	KGW on OPT-1.3B: 100% → 0%.	Not evaluated.
Watermark	Multi-key/API attacks (Pang et al., 2024)	ASR; ACC	Detection-API attack: removal ASR = 1.00 and spoofing ASR = 0.98 for KGW.	DP-inspired API defense: spoofing ASR 0.98 → 0; ACC 0.982 → 0.972.
Watermark	SCTS (Wu & Chandrasekaran, 2024)	AUROC	UMD ($c = 4$, left hash): 0.9884 → 0.1332.	Not proposed.
Watermark	Oracle-guided removal (Zhang et al., 2024a)	z-score; p -value	Unigram: z-score 8.210 → 1.456; p -value 4.563e-11 → 0.208.	Not proposed.
Watermark	CWRA (He et al., 2024)	TPR@10%FPR; AUROC	KGW TPR 0.992 → 0.213 (a 0.779 absolute decrease).	X-SIR: mean TPR +0.448 and AUROC +0.221 in cross-lingual settings.
Watermark	CT ² de-watermarking (Chakraborty et al., 2023)	De-watermarking accuracy	Across the tested paraphraser, the maximum is 99.2%; GPT-3.5 reaches 99.1%.	Not proposed.
Paraphrase	DIPPER (Krishna et al., 2023)	Detection rate at 1% FPR	DetectGPT on GPT-2-XL: 70.3% → 4.6% (−65.7 pp).	BM25 retrieval detects 97.8% of the strongest GPT-2-XL paraphrases at 1% FPR.
Paraphrase	Recursive paraphrasing (Sadasivan et al., 2025)	TPR@1%FPR	Watermark detector: 99.3% → 9.7% (−89.6 pp).	Not proposed.
Paraphrase	ChatGPT polishing (Yang et al., 2023)	ACC	GPTZero on CDB: 0.4406.	RoBERTa-HPPT on CDB: 0.8825.
Paraphrase	Expert editing (Beemo) (Artemova et al., 2025)	AUROC	Expert editing reduces detector AUROC by up to about 22 pp.	Not proposed.
Paraphrase	Iterative rewriting (Tripto et al., 2024)	Macro F1	Fine-tuned BERT on ELI5: 0.96 → 0.51 after ChatGPT paraphrasing.	Not proposed.
Paraphrase	TH-Bench (HMGC result) (Zheng et al., 2025)	AUROC	Log-Likelihood detector on Reuters: 0.808 → 0.020.	Not proposed.
Paraphrase	Commercial humanizers (Masrouf et al., 2025)	TPR@5%FPR	Binoculars: 0.9415 → 0.2823 after humanization.	DAMAGE: TPR = 0.9826 on humanized academic text.
Paraphrase	AuthorMist (David & Gervais, 2025)	ASR	Best cross-detector model: mean ASR = 95.17%.	Not proposed.
Paraphrase	ESPERANTO back-translation (Ayoobi et al., 2025)	TPR@1%FPR	GPTZero on R-L: 0.65 → 0.09.	MESAS (Uni + Bi): mean post-attack TPR = 0.947.
Prompt	EScaPe (Kumarage et al., 2023)	F1	Mean OpenAI-FT F1 across tasks/models: 0.954 → 0.537.	Not proposed.
Prompt	FAILOpt (Park et al., 2024)	AUROC	ChatGPT detector on SQuAD: 0.9184 → 0.4452.	Augmented detector: 0.4452 → 0.9888.
Prompt	SICO (Lu et al., 2024)	AUROC	Log-Rank on writing: 0.914 → 0.147 (−0.767).	Ensemble on writing/SICO-Gen: 0.099 → 0.691 (+0.592).
Prompt	OUTFOX (Koike et al., 2024)	F1	HC3 detector: 0.738 → 0.007 (−73.1 pp).	Attack-aware OUTFOX detector: 0.394 → 0.807 (+41.3 pp).
Prompt	Personalized generation (Gao et al., 2026)	AUROC; ACC	Mean AUROC = 0.3233 on Stylo-Literary; Fast-DetectGPT = 0.0871.	Mixed-domain RoBERTa: ACC = 99.95% on Stylo-Literary and 71.94% on Stylo-Blog.
Adversarial text	Query-based substitutions (Shi et al., 2024)	AUROC	DetectGPT on GPT-2-XL/XSum: 0.844 → 0.039.	Not proposed.
Adversarial text	Editing attacks (Wang et al., 2024b)	AUROC	Typo insertion against DetectGPT-10z: 0.8516 → 0.2735.	Anomaly-filter patch: 0.2735 → 0.6032 (mean gain 0.2285 across variants).
Adversarial text	RAFT (Wang et al., 2024a)	AUROC; TPR@5%FPR	Fast-DetectGPT on XSum: AUROC 0.9903 → 0.0006; TPR 0.94 → 0.	Adversarially trained Raidar on XSum: AUROC 0.6000 → 0.7333.
Adversarial text	SilverSpeak (Creo & Pudasaini, 2025)	MCC	Mean MCC: 0.64 → −0.01.	Not evaluated.
Adversarial text	PDFuzz (Creo, 2025)	ACC	ArguGPT: 0.936 → 0.504.	Not evaluated.
Adversarial text	Mixed hierarchical perturbations (PIFE evaluation) (Teja et al., 2025)	TPR@1%FPR	Adversarially trained ModernBERT: 0.488.	PIFE under the same mixed attacks: 0.826.

Notes: Arrows report the original before→after values; “pp” denotes percentage points. “Not proposed” means that the cited study did not propose a defense method; “Not evaluated” means that mitigation ideas were discussed without an empirical defense result. Attack and defense values within a row may originate from different detector configurations reported in the same study and should not necessarily be interpreted as a matched before-and-after comparison. FNR: false-negative rate; FNR*: FNR restricted to quality-preserving paraphrases; ASR: attack success rate; TPR@ X %FPR: true-positive rate at an X % false-positive rate; AUROC: area under the receiver-operating-characteristic curve; ACC: accuracy; MCC: Matthews correlation coefficient.

reduces KGW's TPR from 100% to 0% (Chang et al., 2025), while stealing-assisted scrubbing raises the FNR of KGW2-SelfHash from 0.00 to 0.79 on Reddit Writing and from 0.02 to 0.90 on Dolly-CW (Jovanović et al., 2024). The detection-API attacks of Pang et al. (2024) reach removal and spoofing ASRs of 1.00 and 0.98, respectively. Their differential-privacy-inspired API countermeasure drives spoofing ASR to 0 while changing detector accuracy from 0.982 to 0.972, although the same guarantee does not extend to removal attacks. SCTS (Wu & Chandrasekaran, 2024) and oracle-guided removal (Zhang et al., 2024a) report strong attacks but do not empirically evaluate a countermeasure in the cited studies.

Paraphrasing-based attacks likewise expose substantial weaknesses. DIPPER reduces DetectGPT's detection rate on GPT-2-XL from 70.3% to 4.6% at a 1% FPR, whereas retrieval-based attribution detects 97.8% of the strongest paraphrases in that setting (Krishna et al., 2023). Recursive paraphrasing reduces a watermark detector's TPR from 99.3% to 9.7% (Sadasivan et al., 2025). The newly included studies broaden this evidence to commercial humanizers, author-mimicking rewrites, back-translation, and benchmarked human-machine coauthoring: TH-Bench records an AUROC decline from 0.808 to 0.020 for Log-Likelihood on Reuters (Zheng et al., 2025); DAMAGE and MESAS retain TPRs of 0.9826 and 0.947 in their respective evaluated settings (Ayoobi et al., 2025; Masroui et al., 2025); and AuthorMist reports a mean ASR of 95.17% for its best cross-detector model (David & Gervais, 2025). These values should not be treated as a single ranking because their datasets and operating points differ.

Prompt-based adversarial attacks manipulate generation so that the output is less detectable without changing detector parameters. EScaPe reduces mean OpenAI-FT F1 from 0.954 to 0.537 (Kumarage et al., 2023); FAILOpt reduces AUROC from 0.9184 to 0.4452 in the reported SQuAD setting and reaches a maximum ASR of 95.72% (Park et al., 2024); SICO reduces Log-Rank AUROC from 0.914 to 0.147 (Lu et al., 2024); and OUTFOX reduces HC3-detector F1 from 0.738 to 0.007 (Koike et al., 2024). Attack-aware training or ensembling can recover performance in matched settings: the corresponding values rise to 0.9888 for FAILOpt, 0.691 for SICO, and 0.807 for OUTFOX. Personalized generation reduces mean AUROC to 0.3233 on Stylo-Literary, while mixed-domain RoBERTa fine-tuning reaches 99.95% accuracy on Stylo-Literary but only 71.94% on Stylo-Blog (Gao et al., 2026).

Adversarial-text attacks introduce strategically selected character-, token-, or sentence-level perturbations. Query-based substitutions reduce DetectGPT AUROC on GPT-2-XL/XSum from 0.844 to 0.039 (Shi et al., 2024), and typo insertion reduces DetectGPT-10z AUROC from 0.8516 to 0.2735; the anomaly-filter patch recovers it to 0.6032 (Wang et al., 2024b). RAFT reduces Fast-DetectGPT AUROC on XSum from 0.9903 to 0.0006 and TPR at 5% FPR from 0.94 to 0, while adversarial training raises Raidar AUROC from 0.6000 to 0.7333 (Wang et al., 2024a). The newly added SilverSpeak study demonstrates degradation through Unicode homoglyph substitutions, whereas PDFuzz exploits discrepancies between visible PDF rendering and extracted character order (Creo, 2025; Creo & Pudasaini, 2025); PIFE reports improved robustness over standard adversarial training under sentence-level mixed attacks, raising TPR at 1% FPR from 0.488 to 0.826 and AUROC from 0.921 to 0.974 (Teja et al., 2025). Such gains remain tied to the evaluated attack distribution and therefore do not establish universal robustness.

7. Discussion

7.1. Watermark attacks and defenses

The evidence synthesized in this review suggests that watermark robustness should be treated as a system-level property rather than as an intrinsic property of the embedded signal. A watermark can fail at several distinct points: an attacker may infer the generation rule from repeated outputs, exploit information exposed by a public detector, remove the signal through local or global text transformation, or

transfer the content into another linguistic representation. These attack surfaces are not interchangeable. Watermark stealing and multi-key or detector-API attacks exploit information leakage from the generation or verification interface (Jovanović et al., 2024; Pang et al., 2024), whereas smoothing, SCTS, oracle-guided removal, and CT² directly reduce the statistical evidence retained in the text (Chakraborty et al., 2023; Chang et al., 2025; Wu & Chandrasekaran, 2024; Zhang et al., 2024a). Cross-lingual removal further shows that a signal stable within one language may not survive changes in tokenization, syntax, and lexical choice after translation (He et al., 2024). Consequently, resistance to one paraphraser or substitution strategy does not imply resistance to an adaptive attacker with a different interface or transformation budget.

The defense results in Table 2 reinforce this asymmetry between attacks and countermeasures. Of the seven watermark-attack settings summarized in the table, only the detector-API defense and X-SIR report an empirically evaluated recovery effect; two studies discuss possible mitigations without evaluating their defense effect, and three do not propose a defense. The differential-privacy-inspired API mechanism substantially reduces spoofing by perturbing exposed detector scores, but it is less effective against removal attacks because the attacker does not need to reconstruct the watermark distribution (Pang et al., 2024). X-SIR instead improves cross-lingual consistency by aligning the watermark with multilingual semantic structure (He et al., 2024). This design addresses translation-based removal, but it does not directly prevent watermark stealing, confidence-guided smoothing, or oracle-guided editing. Similarly, semantics-based schemes such as SemaMark improve resilience to paraphrasing by binding token selection to semantic representations (Ren et al., 2024), yet semantic invariance alone does not remove the risks created by repeated queries, public confidence scores, or compromised keys.

These findings also expose an important limitation in how watermark robustness is currently evaluated. Reported studies use different watermark generators, language models, datasets, text lengths, attack budgets, quality constraints, and operating points. A large decrease in TPR under one configuration therefore cannot be interpreted as a universal ranking of watermark schemes, and a defense gain against a matched attack may reflect specialization to that threat model. More informative evaluation would hold text quality and false-positive constraints constant while varying the attacker's knowledge, query access, transformation budget, target language, and ability to observe detector scores. It should also report both removal and spoofing outcomes, because a mechanism that makes removal more difficult may still permit false attribution. Evaluating adaptive compositions—for example, watermark stealing followed by cross-lingual paraphrasing or confidence-guided token substitution—would provide a more realistic test than assessing each attack in isolation.

From a deployment perspective, the most defensible strategy is therefore layered rather than algorithm-specific. Signal-level robustness should be combined with key rotation and separation, restricted detector feedback, rate limiting, input-dependent score randomization, and monitoring for repeated adaptive queries. Where provenance records can be retained lawfully, retrieval or cryptographic attribution may provide complementary evidence when the statistical watermark has been weakened. However, each layer introduces costs or constraints: limiting detector transparency can reduce auditability, key rotation complicates verification, multilingual semantic schemes depend on language coverage, and retaining generated outputs raises privacy and storage concerns. Future watermark systems should consequently be evaluated not only by clean-text detectability, but also by their security under adaptive access, utility after attack, false-attribution risk, multilingual transfer, and operational cost. This broader perspective explains why recent watermark research is central to the adversarial detection problem: watermarking is not merely a detection technique, but an interactive security mechanism whose reliability depends on the surrounding generation and verification system.

7.2. Paraphrasing attacks and defenses

Paraphrasing attacks reveal a fundamental mismatch between what many MGTD systems measure and what authorship attribution requires. Likelihood- and classifier-based detectors often learn lexical, syntactic, or stylistic regularities associated with a generator, whereas paraphrasing deliberately changes these regularities while preserving most of the underlying meaning. The reviewed studies show that this transformation is better understood as a continuum than as a single attack: it ranges from one-pass model rewriting and recursive paraphrasing to expert editing, commercial humanization, multilingual back-translation, and author-style imitation (Artemova et al., 2025; Ayooobi et al., 2025; David & Gervais, 2025; Krishna et al., 2023; Masrour et al., 2025). Robustness to one point on this continuum does not guarantee robustness elsewhere. A detector trained on the artifacts of an automated paraphraser may fail when an expert removes those artifacts, while a detector adapted to one commercial humanizer may learn tool-specific regularities rather than a stable signature of machine involvement.

The strongest defenses change either the evidence source or the training distribution. Retrieval-based attribution avoids relying exclusively on distributional cues by matching a submitted passage to stored model outputs (Krishna et al., 2023). This can remain effective after substantial rewriting, but only when the source output is present in the retrieval corpus; its deployment also requires decisions about data retention, privacy, indexing cost, and access control. Attack-aware training instead exposes a classifier to polished, back-translated, or humanized samples. RoBERTa-HPPT, DAMAGE, and MESAS demonstrate that this strategy can recover strong performance within their evaluated settings (Ayooobi et al., 2025; Masrour et al., 2025; Yang et al., 2023). The remaining question is whether the recovered performance represents a learned invariance or recognition of attack-specific traces. DAMAGE's evaluation on unseen humanizers provides evidence of cross-tool generalization, whereas the continued difficulty of expert editing in Beemo indicates that human revision may remove generator cues without introducing a consistent alternative signature (Artemova et al., 2025).

Recent benchmarks also change the interpretation of attack success. TH-Bench evaluates evasion effectiveness jointly with text quality and computational overhead and finds that no evaluated humanization attack dominates all three dimensions (Zheng et al., 2025). Beemo moves beyond the conventional human-versus-machine setting by including expert-edited and LLM-edited outputs, thereby representing multi-author workflows that are common in practice (Artemova et al., 2025). These resources suggest that detector failure should not be analyzed independently of editing intensity, semantic preservation, and authorship composition. Future defenses should therefore be evaluated on a graded authorship task—fully generated, model-edited, human-edited, and collaboratively written—rather than only on a binary label. This change would make the evaluation target more consistent with the actual phenomenon being detected.

7.3. Prompt-based attacks and defenses

Prompt-based attacks differ from post-generation rewriting because they alter the distribution at the point of generation. EScaPe, FAILOpt, SICO, and OUTFOX guide the model toward outputs that avoid detector-sensitive features before a detector observes the text (Koike et al., 2024; Kumarage et al., 2023; Lu et al., 2024; Park et al., 2024). The resulting samples need not contain the editing artifacts created by a paraphraser, which limits the transferability of defenses based on post hoc normalization or rewrite detection. Prompt optimization can also exploit correlations specific to a detector, dataset, or writing task. High attack success in a matched setting may therefore reflect a narrow shortcut in the target detector rather than a general property of machine-generated text.

The reviewed defenses mainly use data augmentation, detector ensembles, or attack-aware in-context examples. Their reported recov-

eries show that detector adaptation can be effective when the attack family is represented during defense construction. However, this creates a moving-target problem: retraining against one optimized prompt may shift the attack toward another style, task, or model. Ensemble methods reduce dependence on a single score but remain vulnerable when their components share training data or similar likelihood-based features. OUTFOX offers a more dynamic alternative by adding adversarial examples to the detector context at inference time (Koike et al., 2024), although this approach increases prompt length and inference cost and remains dependent on the relevance of the selected examples.

Personalized generation broadens this problem beyond intentionally adversarial prompts. StyloBench shows that imitating an individual's literary or blog style can invert features that separate human and machine text in general-domain data (Gao et al., 2026). This finding suggests that a detector can fail even when the user's objective is faithful personalization rather than evasion. Mixed-domain fine-tuning can recover performance on a represented personalization domain, but the weaker transfer to another domain indicates that domain coverage remains a central constraint. Prompt-robust detection should therefore be evaluated under unseen instructions, generators, tasks, and personal styles. A useful defense must distinguish stable evidence of machine involvement from features that merely correlate with a particular prompting convention.

7.4. Adversarial-text attacks and defenses

Adversarial-text attacks demonstrate that the MGTD attack surface extends beyond semantic rewriting. Query-based substitutions and RAFT manipulate selected words, SilverSpeak exploits visually similar characters, and PDFuzz targets the discrepancy between a document's visible layout and its extracted character sequence (Creo, 2025; Creo & Pudasaini, 2025; Shi et al., 2024; Wang et al., 2024a). These attacks operate at different layers of the detection pipeline: lexical representation, Unicode processing, tokenization, document parsing, and classification. A detector may therefore fail even when the visible wording and human interpretation change little. This observation is important for benchmark design because robustness measured on plain-text inputs does not establish robustness after real document ingestion.

Existing defenses reflect the same layer dependence. Anomaly filtering can reduce the influence of suspicious tokens, adversarial training can improve robustness to represented perturbations, and normalization can remove predictable character- or formatting-level variation (Wang et al., 2024a,b). PIFE goes further by measuring the discrepancy between an input and its normalized representation and using the transformation itself as detection evidence (Teja et al., 2025). Its advantage over standard adversarial training under mixed sentence-level attacks suggests that explicitly modeling perturbation magnitude can be more transferable than memorizing individual attack samples. Nevertheless, an adaptive attacker may search for transformations that survive the chosen normalization pipeline or mimic naturally occurring editing behavior. Strong normalization may also remove legitimate stylistic variation, creating false positives for non-native writers, accessibility tools, or unusual document encodings.

These results support a pipeline-oriented defense strategy. Character canonicalization, script and homoglyph checks, parser consistency tests, token-level anomaly detection, semantic comparison, and adversarially trained classification address different failure modes and should be evaluated both separately and in combination. Such a pipeline must preserve an auditable record of which stage triggered the decision; otherwise, improved aggregate accuracy may conceal systematic errors affecting particular languages or writing communities. Evaluation should include adaptive attacks against the complete preprocessing and classification stack, not only attacks constructed against the final classifier.

Table 3

Representative benchmarks and evaluation frameworks for adversarial machine-generated-text detection. The table distinguishes general evaluation resources from targeted attack–defense frameworks and summarizes the boundary of the conclusions supported by each resource.

Resource	Type	Coverage	Primary evaluation contribution	Interpretive boundary
TH-Bench (Zheng et al., 2025)	Multi-attack benchmark	Six evasion attacks, 13 detectors, six datasets, 19 domains, and outputs from 11 LLMs	Jointly compares evasion effectiveness, text quality, and computational overhead	Covers paraphrasing, perturbation, and data-mixing attacks rather than the full watermark, prompt, and document-layer attack space
Beemo (Artemova et al., 2025)	Multi-author benchmark	19.6k human-written, machine-generated, expert-edited, and LLM-edited texts; ten instruction-tuned LLMs; 33 detector configurations	Tests expert and model editing, edit types, and out-of-domain detection in realistic multi-author settings	Represents selected use cases and editing protocols rather than an adaptive attacker optimizing directly against each detector
DAMAGE (Masrouf et al., 2025)	Targeted attack–defense evaluation	Nineteen commercial humanizer and paraphrasing tools evaluated against open-source and commercial detectors	Tests cross-humanizer generalization and detector-specific adversarial fine-tuning	Conclusions concern humanization-style rewriting and do not establish robustness to unrelated prompt, watermark, or document attacks
StyloBench (Gao et al., 2026)	Personalization benchmark	Literary and blog texts paired with LLM-generated style imitations	Reveals feature inversion and evaluates robustness under personalized generation	Covers two personalization scenarios; transfer to other genres, languages, and individual styles remains uncertain
RAFT (Wang et al., 2024a)	Targeted black-box attack framework	Grammar-aware word substitutions across multiple detectors, domains, and source models, with human quality evaluation	Tests realistic transfer attacks and the benefit of adversarial training	Evaluates one substitution mechanism and assumes access to an auxiliary model's embedding representation
PIFE (Teja et al., 2025)	Hierarchical robustness framework	Character-, word-, and sentence-level attacks evaluated at strict low-FPR operating points	Compares standard adversarial training with perturbation-aware normalization features	Results depend on the evaluated normalization pipeline and attack taxonomy rather than a shared public multi-detector benchmark

7.5. Benchmark coverage and cross-cutting research gaps

The representative resources in Table 3 illustrate why cross-paper comparison remains difficult. Table 3 re-characterizes selected studies from Table 2 as benchmarks or evaluation frameworks: whereas Table 2 reports their quantitative attack and defense effects, Table 3 compares their evaluation coverage, primary contributions, and interpretive boundaries. TH-Bench provides broad multi-attack coverage and explicitly measures attack effectiveness, text quality, and overhead, whereas Beemo and StyloBench focus on expert editing and personalization, respectively (Artemova et al., 2025; Gao et al., 2026; Zheng et al., 2025). DAMAGE, RAFT, and PIFE provide more targeted attack–defense evaluations rather than general-purpose benchmarks (Masrouf et al., 2025; Teja et al., 2025; Wang et al., 2024a). Each resource therefore answers a different question. Combining their headline scores into a single ranking would obscure differences in detector access, text source, transformation budget, quality control, and operating point.

Across these resources, four research gaps are especially persistent. First, most evaluations isolate one attack family, even though practical evasion can compose prompt manipulation, rewriting, character substitution, translation, and document-layer perturbation. Defenses validated against a single transformation may therefore overestimate robustness. Second, generalization is usually tested along only one axis—for example, an unseen humanizer, model, domain, or language—rather than across all of them simultaneously. Third, binary authorship labels do not adequately represent expert-edited, personalized, or human–AI collaborative writing. The intended detection target must be defined more precisely before performance differences can be interpreted as errors. Fourth, reporting remains inconsistent: AUROC, accuracy, F1, ASR, and TPR at different FPRs are often measured on different datasets and

attack budgets. At minimum, future benchmarks should report low-FPR performance, attack success, semantic and factual preservation, human-rated quality, computational overhead, and the detector's access assumptions.

These gaps have direct implications for claims of defense effectiveness. A defense should not be described as robust solely because it restores performance against the attack used during its construction. Stronger evidence would require preregistered or hidden attacks, held-out generators and domains, multilingual and multi-author samples, and adaptive evaluation against the complete deployed pipeline. Operational efficiency constitutes a related but distinct evidence gap and is discussed separately below.

7.6. Practical deployment and efficiency considerations

Practical MGTD requires more than high robustness under a fixed benchmark. A deployable method must produce decisions within the latency, throughput, storage, privacy, and maintenance constraints of its intended setting. These constraints differ substantially between real-time content moderation, large-scale platform screening, offline forensic analysis, and high-stakes academic or regulatory review. A computationally expensive method may remain appropriate for selective verification but may be unsuitable as the first stage of a high-volume screening pipeline. Conversely, a lightweight detector may support rapid triage but require a more reliable secondary analysis before an adverse decision is made. Efficiency should therefore be interpreted relative to the operational role of the detector rather than as an isolated runtime value.

The main detection paradigms distribute their costs differently. Training-based detectors require labeled data, model training, and periodic updating as generators and writing domains change, but deploy-

ment can often be reduced to a single classifier inference once the model has been trained. Their principal operational risk is maintenance: robustness gained through attack-aware training may decay as new generators, prompts, and humanization tools emerge. Zero-shot detectors avoid task-specific retraining but can incur higher per-document latency when they repeatedly perturb an input or query one or more language models. Fast-DetectGPT reduces this burden by replacing repeated perturbation scoring with conditional probability curvature, while Bayesian surrogate modeling reduces the number of expensive detector queries (Bao et al., 2024; Miao et al., 2024). Lightweight tools such as HawkGPT indicate that simplified statistical signals may be suitable for web deployment, although such efficiency does not by itself establish robustness to adaptive evasion (Vasilatos et al., 2025).

Watermark-based detection shifts part of the cost from post hoc classification to generation and verification. Deployment requires access to the watermarking process, secure key or model management, and a verification interface. Public confidence scores can facilitate auditing but also expose information that enables adaptive removal or spoofing. Restricting score precision, rate limiting, authentication, and input-dependent score randomization may reduce this risk, but they introduce additional service logic and monitoring requirements (Pang et al., 2024). Key rotation and multilingual watermarking further increase operational complexity because verification must remain consistent across model versions, languages, and distributed services. Thus, a watermark detector that is statistically efficient may still have substantial system-level deployment costs.

Defense mechanisms introduce their own resource trade-offs. Retrieval-based attribution requires providers to retain, index, and search generated outputs; semantic retrieval can improve resilience to paraphrasing but increases storage, indexing, and query costs and raises privacy and access-control concerns (Krishna et al., 2023). Attack-aware training and adversarial training move much of the expense offline, but they require repeated collection and generation of attack samples and periodic model updating as attacks evolve (Masrour et al., 2025; Wang et al., 2024a). Detector ensembles multiply inference and memory requirements, while attack-aware in-context detection increases prompt length and model-inference cost (Koike et al., 2024; Lu et al., 2024). Multi-stage normalization and perturbation-aware methods such as PIFE add preprocessing and feature-computation stages before classification (Teja et al., 2025). Their reported robustness gains must therefore be weighed against the accumulated latency of the complete pipeline rather than only the runtime of the final classifier.

The present literature does not support a reliable numerical efficiency ranking because most studies do not report results under common hardware, batch size, text length, model-access, or caching conditions. TH-Bench is a useful step toward a broader evaluation protocol because it evaluates computational overhead together with evasion effectiveness and text quality (Zheng et al., 2025). Future benchmarks should extend this practice by reporting per-document latency, throughput, peak memory, hardware and model specifications, the number of model or API calls, preprocessing time, storage and index size, and the frequency and cost of retraining. For defenses that rely on external services, monetary query cost and failure behavior under rate limits should also be reported. Jointly measuring these quantities with low-FPR detection performance and post-attack robustness would make it possible to distinguish methods that are merely accurate in an experimental setting from those that can be maintained reliably at deployment scale.

8. Conclusion and future research directions

This review organizes attacks against machine-generated text detection into four categories: watermark, paraphrasing, prompt-based, and adversarial-text attacks. Across these categories, the reviewed evidence shows that detectors remain vulnerable when the generator, domain, language, editing process, or input representation changes. Existing defenses can recover performance in matched settings, but such gains do

not establish general resistance to unseen or adaptive attacks. MGTD should therefore be evaluated as an evolving detection pipeline rather than as a fixed binary classifier.

Several limitations constrain the interpretation of the current evidence. Cross-paper comparisons are confounded by differences in datasets, generators, attack budgets, quality constraints, metrics, and false-positive operating points. Most defenses are evaluated against a single attack family, while adaptive attack compositions and complete preprocessing pipelines receive limited attention. Binary authorship labels also underrepresent expert-edited, personalized, and human-AI collaborative writing. In addition, low-FPR performance, multilingual transfer, latency, throughput, memory use, query cost, storage requirements, and retraining costs are reported inconsistently. Reported robustness should consequently be interpreted within each study's threat model and deployment setting.

These limitations suggest three priorities for future research:

1. **Standardized evaluation under adaptive and compound attacks.** Future benchmarks should combine multiple attack surfaces and include held-out generators, domains, languages, and attack strategies. Evaluation should control text quality and false-positive constraints while reporting low-FPR detection, semantic and factual preservation, attack success, and computational overhead.
2. **Graded modeling of human-AI collaborative writing.** Future detectors should move beyond a single human-versus-machine label. They should quantify the extent and location of model involvement, represent uncertainty, and evaluate expert-edited, personalized, and collaboratively written text.
3. **Layered and deployable defense pipelines.** Robust systems should combine complementary evidence from provenance, watermarking, retrieval, normalization, anomaly detection, and attack-aware classification. These pipelines should be evaluated end to end for adaptive robustness, privacy, fairness, maintainability, and operational efficiency.

Progress in MGTD will depend less on optimizing a single benchmark score and more on demonstrating robustness under explicit and realistic constraints. This perspective reframes MGTD as a systems-level security and provenance problem involving generation, transformation, verification, and deployment.

CRedit authorship contribution statement

Deyu Meng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization; **Tad Gonsalves:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No new datasets were generated or analyzed in this review. The categorized literature, paper links, and identified research repositories are available at <https://github.com/AIGC1999/A-Survey-of-Anti-Detection-Techniques-for-Machine-Generated-Texts>.

Supplementary material

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References

- Artemova, E., Lucas, J. S., Venkatraman, S., Lee, J., Tilga, S., Uchendu, A., & Mikhailov, V. (2025). Beemo: Benchmark of expert-edited machine-generated outputs. In L. Chiruzzo, A. Ritter, & L. Wang (Eds.), *Proceedings of the 2025 conference of the nations of the Americas chapter of the association for computational linguistics: Human language technologies (Volume 1: Long papers)* (pp. 6992–7018). Albuquerque, New Mexico: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.naacl-long.357>
- Ayoobi, N., Knab, L., Cheng, W., Pantoja, D., Alikhani, H., Flamant, S., Kim, J., & Mukherjee, A. (2025). ESPERANTO: Evaluating synthesized phrases to enhance robustness in AI detection for text origination. In *Proceedings of the 36th ACM conference on hypertext and social media* (pp. 1–10). Association for Computing Machinery. <https://doi.org/10.1145/3720553.3746665>
- Bakhtin, A., Gross, S., Ott, M., Deng, Y., Ranzato, M., & Szlam, A. (2019). Real or fake? Learning to discriminate machine from human generated text. <https://arxiv.org/abs/1906.03351>
- Bao, G., Zhao, Y., Teng, Z., Yang, L., & Zhang, Y. (2024). Fast-detectGPT: Efficient zero-shot detection of machine-generated text via conditional probability curvature. <https://arxiv.org/abs/2310.05130>
- Bhattacharjee, A., Kumara, T., Moraffah, R., & Liu, H. (2023). ConDA: Contrastive domain adaptation for AI-generated text detection. In J. C. Park, Y. Arase, B. Hu, W. Lu, D. Wijaya, A. Purwaranti, & A. A. Krishnadh (Eds.), *Proceedings of the 13th international joint conference on natural language processing and the 3rd conference of the Asia-Pacific chapter of the association for computational linguistics (Volume 1: Long papers)* (pp. 598–610). Nusa Dua, Bali: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.ijcnlp-main.40>
- Chakraborty, M., Tomney, S. M. T. I., Zaman, S. M. M., Gautam, S., Kumar, T., Sharma, K., Barman, N., Gupta, C., Jain, V., Chadha, A., Sheth, A., & Das, A. (2023). Counter turing test (CT2): AI-generated text detection is not as easy as you may think - introducing AI detectability index (ADI). In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 2206–2239). Singapore: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.emnlp-main.136>
- Chang, H., Hassani, H., & Shokri, R. (2025). Watermark smoothing attacks against language models. <https://arxiv.org/abs/2407.14206>
- Chen, W., Wang, Q., Long, Z., Zhang, X., Lu, Z., Li, B., Wang, S., Xu, J., Bai, X., Huang, X., & Wei, Z. (2023a). DISC-FinLLM: A Chinese financial large language model based on multiple experts fine-tuning. <https://arxiv.org/abs/2310.15205>
- Chen, Y., Kang, H., Zhai, V., Li, L., Singh, R., & Raj, B. (2023b). GPT-sentinel: Distinguishing human and ChatGPT generated content. <https://arxiv.org/abs/2305.07969>
- Creo, A. (2025). Complete evasion, zero modification: PDF attacks on AI text detection. <https://arxiv.org/abs/2508.01887>
- Creo, A., & Pudasaini, S. (2025). SilverSpeak: Evading AI-generated text detectors using homoglyphs. In *Proceedings of the 1st workshop on genAI content detection (genAIDetect)* (pp. 1–46). Abu Dhabi, UAE: International Conference on Computational Linguistics. <https://aclanthology.org/2025.genaidetect-1.1/>
- Dan, Y., Lei, Z., Gu, Y., Li, Y., Yin, J., Lin, J., Ye, L., Tie, Z., Zhou, Y., Wang, Y., Zhou, A., Zhou, Z., Chen, Q., Zhou, J., He, L., & Qiu, X. (2025). EduChat: A large language model-based conversational agent for intelligent education. In B. Xu, B. Qin, Y. Tzitzikas, Z. Wang, X. L. Mao, M. Liu, P. Fafalios, Y. Liu, Z. Tian, & M. Mountantonakis (Eds.), *China conference on knowledge graph and semantic computing and international joint conference on knowledge graphs* (pp. 297–308). Singapore: Springer Nature Singapore.
- David, I., & Gervais, A. (2025). AuthorMist: Evading AI text detectors with reinforcement learning. <https://arxiv.org/abs/2503.08716>
- DeepSeek-AI, Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., Liu, A., Xue, B., Wang, B., Wu, B., Feng, B., Lu, C., Zhao, C., Deng, C., Zhang, C., Ruan, C., Dai, D., Chen, D., Ji, D., Li, E., Lin, F., Dai, F., Luo, F., Hao, G., Chen, G., Li, G., Zhang, H., Bao, H., Xu, H., Wang, H., Ding, H., Xin, H., Gao, H., Qu, H., Li, H., Guo, J., Li, J., Wang, J., Chen, J., Yuan, J., Qiu, J., Li, J., Cai, J. L., Ni, J., Liang, J., Chen, J., Dong, K., Hu, K., Gao, K., Guan, K., Huang, K., Yu, K., Wang, L., Zhang, L., Zhao, L., Wang, L., Zhang, L., Xu, L., Xia, L., Zhang, M., Zhang, M., Tang, M., Li, M., Wang, M., Li, M., Tian, N., Huang, P., Zhang, P., Wang, Q., Chen, Q., Du, Q., Ge, R., Zhang, R., Pan, R., Wang, R., Chen, R. J., Jin, R. L., Chen, R., Lu, S., Zhou, S., Chen, S., Ye, S., Wang, S., Yu, S., Zhou, S., Pan, S. et al. (2025). Deepseek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. <https://arxiv.org/abs/2501.12948>
- Eke, D. O. (2023). ChatGPT and the rise of generative AI: Threat to academic integrity? *Journal of Responsible Technology*, 13, 100060. <https://doi.org/https://doi.org/10.1016/j.jrt.2023.100060>
- Emanuela, A. R., Cristina, B. A., & Luminița, S. (2024). AI and prompt engineering: The new weapons of social engineering attacks. In *2024 16th International conference on electronics, computers and artificial intelligence (ECAI)* (pp. 1–6). <https://doi.org/10.1109/ECAI61503.2024.10607415>
- Fagni, T., Falchi, F., Gambini, M., Martella, A., & Tesconi, M. (2021). Tweepfake: About detecting deepfake tweets. *PloS One*, 16(5), e0251415.
- Fernandez, P., Chaffin, A., Tit, K., Chappelier, V., & Furon, T. (2023). Three bricks to consolidate watermarks for large language models. <https://arxiv.org/abs/2308.00113>
- Gao, L., Li, X., Wang, C., Li, M., Liu, W., Song, Z., Zhang, J., Yan, R., Nakov, P., & Chen, X. (2026). When personalization tricks detectors: The feature-inversion trap in machine-generated text detection. In *Proceedings of the 64th annual meeting of the association for computational linguistics (Volume 1: Long papers)* (pp. 43143–43171). San Diego, California, United States: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2026.acl-long.1998>
- Gehrmann, S., Strobelt, H., & Rush, A. (2019). GLTR: Statistical detection and visualization of generated text. In M. R. Costa-jussà, & E. Alfonseca (Eds.), *Proceedings of the 57th annual meeting of the association for computational linguistics: System demonstrations* (pp. 111–116). Florence, Italy: Association for Computational Linguistics. <https://doi.org/10.18653/v1/P19-3019>
- Gu, C., Huang, C., Zheng, X., Chang, K. W., & Hsieh, C.-J. (2023). Watermarking pre-trained language models with backdoor. <https://arxiv.org/abs/2210.07543>
- He, Z., Zhou, B., Hao, H., Liu, A., Wang, X., Tu, Z., Zhang, Z., & Wang, R. (2024). Can watermarks survive translation? On the cross-lingual consistency of text watermark for large language models. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (Volume 1: Long papers)* (pp. 4115–4129). Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.acl-long.226>
- Ippolito, D., Duckworth, D., Callison-Burch, C., & Eck, D. (2020). Automatic detection of generated text is easiest when humans are fooled. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 1808–1822). Online: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2020.acl-main.164>
- Jawahar, G., Abdul-Mageed, M., & Lakshmanan, V. S. L., (2020). Automatic detection of machine generated text: A critical survey. In D. Scott, N. Bel, & C. Zong (Eds.), *Proceedings of the 28th international conference on computational linguistics* (pp. 2296–2309). Barcelona, Spain (Online): International Committee on Computational Linguistics.
- Jovanović, N., Staab, R., & Vechev, M. (2024). Watermark stealing in large language models. In R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, & F. Berkenkamp (Eds.), *Proceedings of the 41st international conference on machine learning* (pp. 22570–22593). PMLR (vol. 235). Proceedings of Machine Learning Research.
- Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., & Goldstein, T. (2023). A watermark for large language models. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of the 40th international conference on machine learning* (pp. 17061–17084). PMLR (vol. 202). Proceedings of Machine Learning Research.
- Koike, R., Kaneko, M., & Okazaki, N. (2024). Outfox: LLM-generated essay detection through in-context learning with adversarially generated examples. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(19), 21258–21266. <https://doi.org/10.1609/aaai.v38i19.30120>
- Krishna, K., Song, Y., Karpinska, M., Wieting, J., & Iyyer, M. (2023). Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in neural information processing systems* (pp. 27469–27500). Curran Associates, Inc. (vol. 36).
- Kumara, T., Sheth, P., Moraffah, R., Garland, J., & Liu, H. (2023). How reliable are AI-generated-text detectors? An assessment framework using evasive soft prompts. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Findings of the association for computational linguistics: EMNLP 2023* (pp. 1337–1349). Singapore: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.findings-emnlp.94>
- Li, J., Guan, Z., Wang, J., Cheung, C. Y., Zheng, Y., Lim, L. L., Lim, C. C., Ruamviboonsuk, P., Raman, R., Corsino, L. et al. (2024a). Integrated image-based deep learning and language models for primary diabetes care. *Nature Medicine*, 30(10), 2886–2896.
- Li, Q., Yang, X., Wang, H., Wang, Q., Liu, L., Wang, J., Zhang, Y., Chu, M., Hu, S., Chen, Y., Shen, Y., Fan, C., Zhang, W., Xu, T., Gu, J., Zheng, J., & Group, G. Z. A. (2024b). From beginner to expert: Modeling medical knowledge into general LLMs. <https://arxiv.org/abs/2312.01040>
- Liao, Y., Jiang, S., Wang, Y., & Wang, Y. (2024). MING-MOE: Enhancing medical multi-task learning in large language models with sparse mixture of low-rank adapter experts. <https://arxiv.org/abs/2404.09027>
- Liu, A., Pan, L., Hu, X., Li, S., Wen, L., King, I., & Yu, P. S. (2024a). An unforgeable publicly verifiable watermark for large language models. <https://arxiv.org/abs/2307.16230>
- Liu, A., Pan, L., Lu, Y., Li, J., Hu, X., Zhang, X., Wen, L., King, I., Xiong, H., & Yu, P. (2024b). A survey of text watermarking in the era of large language models. *ACM Computing Surveys*, 57(2). <https://doi.org/10.1145/3691626>
- Liu, Y., Zhang, Z., Zhang, W., Yue, S., Zhao, X., Cheng, X., Zhang, Y., & Hu, H. (2023). ArguGPT: Evaluating, understanding and identifying argumentative essays generated by GPT models. <https://arxiv.org/abs/2304.07666>
- Lu, N., Liu, S., He, R., Wang, Q., Ong, Y. S., & Tang, K. (2024). Large language models can be guided to evade AI-generated text detection. <https://arxiv.org/abs/2305.10847>
- Luo, L., Ning, J., Zhao, Y., Wang, Z., Ding, Z., Chen, P., Fu, W., Han, Q., Xu, G., Qiu, Y., Pan, D., Li, J., Li, H., Feng, W., Tu, S., Liu, Y., Yang, Z., Wang, J., Sun, Y., & Lin, H. (2024). Taiyi: A bilingual fine-tuned large language model for diverse biomedical tasks. *Journal of the American Medical Informatics Association*, 31(9), 1865–1874. <https://doi.org/10.1093/jamia/ocae037>
- Martínez, S., Castañeda, J. M., & Manrique, R. (2025). A scalable framework for legal text understanding in regulatory and financial contexts. In C.-C. Chen, A. Moreno-Sandoval, J. Huang, Q. Xie, S. Ananiadou, & H.-H. Chen (Eds.), *Proceedings of the joint workshop of the 9th financial technology and natural language processing (finNLP), the 6th financial narrative processing (FNP), and the 1st workshop on large language models for finance and legal (LLMFinLegal)* (pp. 326–334). Abu Dhabi, UAE: Association for Computational Linguistics. [https://aclanthology.org/2025.genaidetect-1.9/](https://aclanthology.org/2025.finllp-1.39/)
- Masrour, E., Emi, B. N., & Spero, M. (2025). DAMAGE: Detecting adversarially modified AI generated text. In *Proceedings of the 1st workshop on genAI content detection (genAIDetect)* (pp. 120–133). Abu Dhabi, UAE: International Conference on Computational Linguistics. <https://aclanthology.org/2025.genaidetect-1.9/>
- Miao, Y., Gao, H., Zhang, H., & Deng, Z. (2024). Efficient detection of LLM-generated texts with a Bayesian surrogate model. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: ACL 2024* (pp. 6118–6130). Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-acl.366>

- Mitchell, E., Lee, Y., Khazatsky, A., Manning, C. D., & Finn, C. (2023). DetectGPT: Zero-shot machine-generated text detection using probability curvature. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of the 40th international conference on machine learning* (pp. 24950–24962). PMLR (vol. 202). Proceedings of Machine Learning Research.
- Ouzzani, M., Hammady, H., Fedorowicz, Z., & Elmagarmid, A. (2016). Rayyan—a web and mobile app for systematic reviews. *Systematic Reviews*, 5(1), 210.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., Stewart, L. A., Thomas, J., Tricco, A. C., Welch, V. A., Whiting, P., & Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372. <https://doi.org/10.1136/bmj.n71>
- Pang, Q., Hu, S., Zheng, W., & Smith, V. (2024). No free lunch in LLM watermarking: Trade-offs in watermarking design choices. In *Proceedings of the 38th international conference on neural information processing systems NIPS '24* (pp. 138756–138788). ISBN 9798331314385 Red Hook, NY, USA: Curran Associates Inc.
- Park, C., Kim, H. J., Kim, J., Kim, Y., Kim, T., Cho, H., Jo, H., Lee, S.-g., & Yoo, K. M. (2024). Investigating the influence of prompt-specific shortcuts in AI generated text detection. <https://arxiv.org/abs/2406.16275>
- Qu, W., Zheng, W., Tao, T., Yin, D., Jiang, Y., Tian, Z., Zou, W., Jia, J., & Zhang, J. (2025). Provably robust multi-bit watermarking for AI-generated text. <https://arxiv.org/abs/2401.16820>
- Ray, P. P. (2023). ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope. *Internet of Things and Cyber-Physical Systems*, 3, 121–154.
- Ren, J., Xu, H., Liu, Y., Cui, Y., Wang, S., Yin, D., & Tang, J. (2024). A robust semantics-based watermark for large language model against paraphrasing. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Findings of the association for computational linguistics: NAACL 2024* (pp. 613–625). Mexico City, Mexico: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-naacl.40>
- Sadasivan, V. S., Kumar, A., Balasubramanian, S., Wang, W., & Feizi, S. (2025). Can AI-generated text be reliably detected? <https://arxiv.org/abs/2303.11156>.
- Shahzad, T., Mazhar, T., Tariq, M. U., Ahmad, W., Ouahada, K., & Hamam, H. (2025). A comprehensive review of large language models: Issues and solutions in learning environments. *Discover Sustainability*, 6(1), 27.
- Shi, Z., Wang, Y., Yin, F., Chen, X., Chang, K. W., & Hsieh, C. J. (2024). Red teaming language model detectors with language models. *Transactions of the Association for Computational Linguistics*, 12, 174–189. <https://aclanthology.org/2024.tacl-1.10/>. https://doi.org/10.1162/tacl_a.00639
- Su, J., Zhuo, T., Wang, D., & Nakov, P. (2023). DetectLLM: Leveraging log rank information for zero-shot detection of machine-generated text. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Findings of the association for computational linguistics: EMNLP 2023* (pp. 12395–12412). Singapore: Association for Computational Linguistics.
- Tang, R., Chuang, Y. N., & Hu, X. (2024). The science of detecting LLM-generated text. *Communications of the ACM*, 67(4), 50–59.
- Teja, L. S., Yadagiri, A., Anish, S. S., Nuthakki, S. G. K., & Pakray, P. (2025). Modeling the attack: Detecting AI-generated text by quantifying adversarial perturbations. <https://arxiv.org/abs/2510.02319>.
- Tian, Y., Chen, H., Wang, X., Bai, Z., Zhang, Q., Li, R., Xu, C., & Wang, Y. (2024). Multiscale positive-unlabeled detection of AI-generated texts. <https://arxiv.org/abs/2305.18149>
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M. A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., & Lample, G. (2023). Llama: Open and efficient foundation language models. <https://arxiv.org/abs/2302.13971>
- Tripto, N. I., Venkatraman, S., Macko, D., Moro, R., Srba, I., Uchendu, A., Le, T., & Lee, D. (2024). A ship of theseus: Curious cases of paraphrasing in LLM-generated texts. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (Volume 1: Long papers)* (pp. 6608–6625). Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.acl-long.357>
- Vasilatos, C., Alam, M., Rahwan, T., Zaki, Y., & Maniatakos, M. (2025). HowkGPT: Investigating the detection of chatGPT-generated university student homework through context-aware perplexity analysis. <https://arxiv.org/abs/2305.18226>.
- Wang, J. L., Li, R., Yang, J., & Mao, C. (2024a). RAFT: Realistic attacks to fool text detectors. In Y. Al-Onaizan, M. Bansal, & Y. N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 16923–16936). Miami, Florida, USA: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.emnlp-main.939>
- Wang, P., Li, L., Ren, K., Jiang, B., Zhang, D., & Qiu, X. (2023). SeqXGPT: Sentence-level AI-generated text detection. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 1144–1156). Singapore: Association for Computational Linguistics. <https://aclanthology.org/2023.emnlp-main.73/>
- Wang, Y., Feng, S., Hou, A., Pu, X., Shen, C., Liu, X., Tsvetkov, Y., & He, T. (2024b). Stumbling blocks: Stress testing the robustness of machine-generated text detectors under attacks. In L. W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (Volume 1: Long papers)* (pp. 2894–2925). Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.acl-long.160>
- Wu, J., Yang, S., Zhan, R., Yuan, Y., Chao, L. S., & Wong, D. F. (2025). A survey on LLM-generated text detection: Necessity, methods, and future directions. *Computational Linguistics*, 51(1), 275–338.
- Wu, Q., & Chandrasekaran, V. (2024). Bypassing LLM watermarks with color-aware substitutions. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (Volume 1: Long papers)* (pp. 8549–8581). Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.acl-long.464>
- Yang, L., Jiang, F., & Li, H. (2023). Is chatGPT involved in texts? Measure the polish ratio to detect chatGPT-generated text. <https://arxiv.org/abs/2307.11380>.
- Yang, Z., Feng, Z., Huo, R., Lin, H., Zheng, H., Nie, R., & Chen, H. (2025). The imitation game revisited: A comprehensive survey on recent advances in AI-generated text detection. *Expert Systems with Applications*, 272, 126694.
- Yue, S., Liu, S., Zhou, Y., Shen, C., Wang, S., Xiao, Y., Li, B., Song, Y., Shen, X., Chen, W., Huang, X., & Wei, Z. (2024). LawLLM: Intelligent legal system with legal reasoning and verifiable retrieval. In M. Onizuka, J.-G. Lee, Y. Tong, C. Xiao, Y. Ishikawa, S. Amer-Yahia, H. V. Jagadish, & K. Lu (Eds.), *Database systems for advanced applications* (pp. 304–321). Singapore: Springer Nature Singapore.
- Zhang, H., Edelman, B. L., Francati, D., Venturi, D., Ateniese, G., & Barak, B. (2024a). Watermarks in the sand: Impossibility of strong watermarking for language models. In R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, & F. Berkenkamp (Eds.), *Proceedings of the 41st international conference on machine learning* (pp. 58851–58880). PMLR (vol. 235). Proceedings of Machine Learning Research.
- Zhang, Z., Qin, W., & Plummer, B. (2024b). Machine-generated text localization. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: ACL 2024* (pp. 8357–8371). Bangkok, Thailand: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-acl.495>
- Zhao, X., Li, L., & Wang, Y. X. (2022). Distillation-resistant watermarking for model protection in NLP. In Y. Goldberg, Z. Kozareva, & Y. Zhang (Eds.), *Findings of the association for computational linguistics: EMNLP 2022* (pp. 5044–5055). Abu Dhabi, United Arab Emirates: Association for Computational Linguistics.
- Zheng, J., Wang, J., Sun, Z., Dong, W., Liu, Y., & He, X. (2025). TH-Bench: Evaluating evading attacks via humanizing AI text on machine-generated text detectors. In *Proceedings of the 31st ACM SIGKDD conference on knowledge discovery and data mining v.2* (pp. 5948–5959). Association for Computing Machinery. <https://doi.org/10.1145/3711896.3737418>